

Spendora

Personal Finance Tracking Application



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General Introduction

In a world where personal financial management is becoming increasingly complex, having effective and adapted tools is essential. Our application, Spendora, addresses this need by offering an intuitive and comprehensive solution to track personal finances, visualize expenses, and achieve savings goals.

This report presents in detail the design, development, and implementation of Spendora, a mobile application developed using Flutter and Python. We detail the project planning, technical architecture, and key features that make Spendora an essential tool for personal financial management.

Through an agile and methodical approach, our team has worked to create an application that not only meets the fundamental needs of financial management but also offers an optimal user experience through relevant graphical visualizations and personalized tracking features.

This document explores the different phases of the project, from needs analysis to final implementation, including the technological choices that guided our development process. We will examine how Spendora addresses the challenges of modern financial management while ensuring data security and user privacy.

The goal of Spendora is to empower users by providing them with clear insights into their spending habits and helping them make informed decisions about their financial future. By combining powerful analytical tools with an accessible interface, we aim to make financial management accessible to everyone, regardless of their level of financial expertise.

Chapter 1

« General Project Framework »

Chapter 1

General Project Framework

1.1 Introduction

Managing personal finances is essential for individuals who want to better organize their income and expenses. This application aims to provide an intuitive and visual solution for tracking finances and achieving savings goals.

1.2 Host Organization

1.2.1 Company Presentation

The application is developed by a team of students as part of an academic project.

1.2.2 Proposed Solutions

- Track income and expenses
- Visualize financial situation using graphs
- Set savings goals and monitor progress
- Generate financial reports

1.2.3 Organization Chart

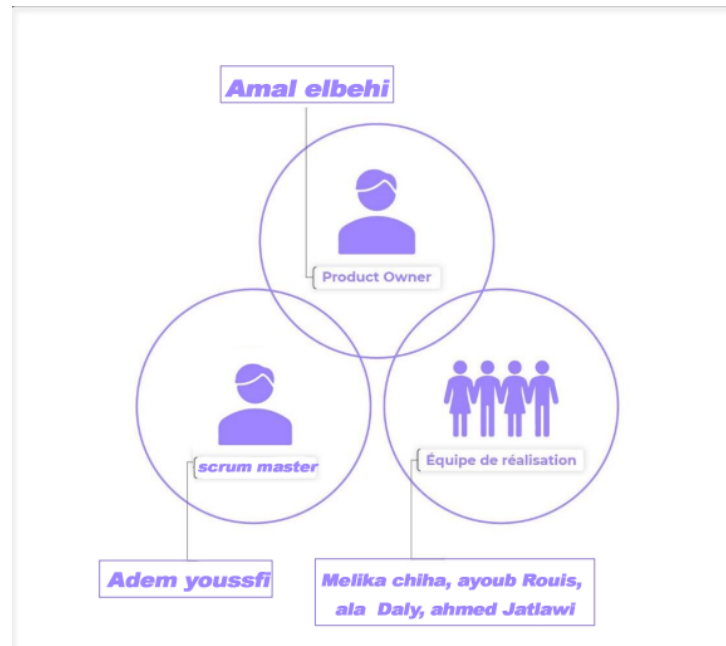


Figure 1.1: Scrum Team and Roles

1.3 Project Framework

1.3.1 Problem Statement

Managing personal finances is essential for individuals who want to better organize their income and expenses.

1.3.2 Existing Solutions Study

In the field of personal financial management, several existing solutions are available:

- **Mint:** A popular application offering expense tracking and budgeting
- **YNAB (You Need A Budget):** A solution focused on envelope budgeting method
- **Personal Capital:** Oriented towards wealth management and investments
- **Excel/Google Sheets:** Basic but flexible solutions for financial tracking

1.3.3 Analysis of Existing Solutions

The analysis of existing solutions reveals several limitations:

- **Complexity:** Many existing applications are too complex for beginner users.
- **Cost:** Most comprehensive solutions are paid with monthly subscriptions.

- **Privacy:** Some applications require access to bank accounts.
- **Customization:** Lack of flexibility in customizing categories and reports.
- **Language:** Few applications are available in French or adapted to the local context.

These limitations justify the development of Spendora, which aims to offer a solution:

- Free and open source
- Easy to use
- Privacy-respecting
- Highly customizable
- Adapted to local needs

1.3.4 Proposed Solution

The main goal is to develop a mobile application that allows users to:

- Track their income and expenses
- Visualize their financial situation
- Set savings goals
- Generate reports

1.4 Development Methodology

1.4.1 Agile Methodologies

We adopted the Scrum methodology for this project.

1.4.2 Choice of SCRUM Methodology

1.4.2.1 Participants in Scrum

- **Product Owner:** Defines needs and specifications
- **Scrum Master:** Ensures proper execution of Scrum
- **Team Members:** Developers, architects, testers

1.4.2.2 Scrum Team Identification

- Adem Youssfi – Developer (Scrum Master)
- Ayoub Rouis – Design
- Ahmed Jatlawi – Design
- Melika Chiha – Design
- Ala Diine Daly – Developer

1.4.3 Modeling Languages

UML diagrams including Use Case and Class diagrams.

1.5 Project Planning

Release	Sprints	Goal	Outcome	Time
Release 1	Sprint 1 & 2	User Management & Transactions	Basic functionality	4 weeks
Release 2	Sprint 3 & 4	Dashboard & Goals	Visualization features	3 weeks
Release 3	Sprint 5 & 6	Reports & Security	Advanced features	2 weeks

Table 1.1: Release Plan

1.6 Conclusion

This chapter has presented a comprehensive overview of the project framework, highlighting several key aspects. We detailed the organizational structure of our team and the methodical approach we've adopted for development. The choice of Scrum methodology has been justified through its flexibility and ability to adapt to changing requirements during the development process. Furthermore, we've outlined our planning strategy, which provides a clear roadmap for project execution while maintaining room for adjustments based on feedback and emerging needs.

The team composition and role distribution have been carefully designed to ensure optimal collaboration and efficient delivery of project milestones. Through this structured approach, we aim to deliver a high-quality financial management solution that meets user expectations while adhering to best practices in software development. The next chapters will build upon this foundation to detail the technical implementation and feature development of Spendora.

Chapter 2

« Project Design and Specification »

Chapter 2

Project Design and Specification

2.1 Introduction

This chapter details the technical specifications and design of the Spendora application.

2.2 Requirements Analysis

2.2.1 Actor Identification

User:

- Registers and logs into the application
- Manages financial transactions
- Views reports and graphs

2.2.2 Functional Requirements

2.2.2.1 Core Features

- User Authentication
- Income/Expense Management
- Dashboard Visualization
- Financial Goals
- Reports and Insights

2.2.2.2 Optional Features

- Dark mode
- Bank API integration
- Multi-currency support

2.2.3 Non-functional Requirements

2.2.3.1 Performance Requirements

- Response time under 2 seconds
- Support for multiple users

2.2.3.2 Security Requirements

- Password encryption
- Protection against attacks
- HTTPS connection

2.2.4 Global Use Case Diagram

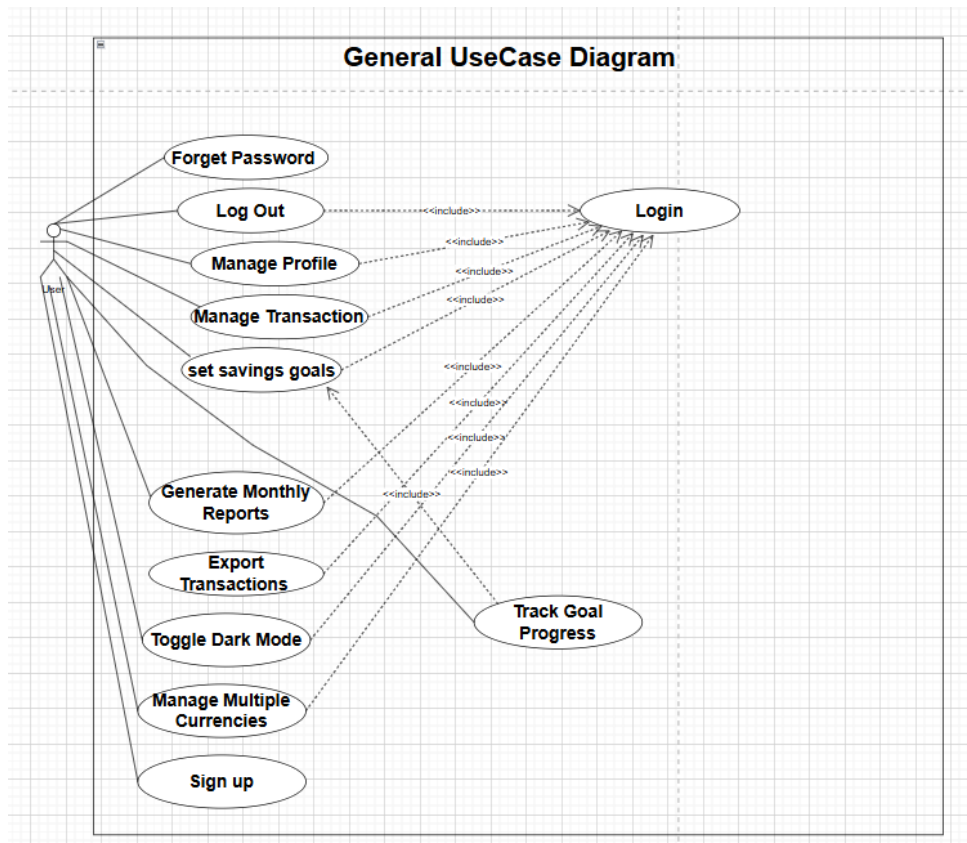


Figure 2.1: System Use Case Diagram

2.2.5 Global Class Diagram

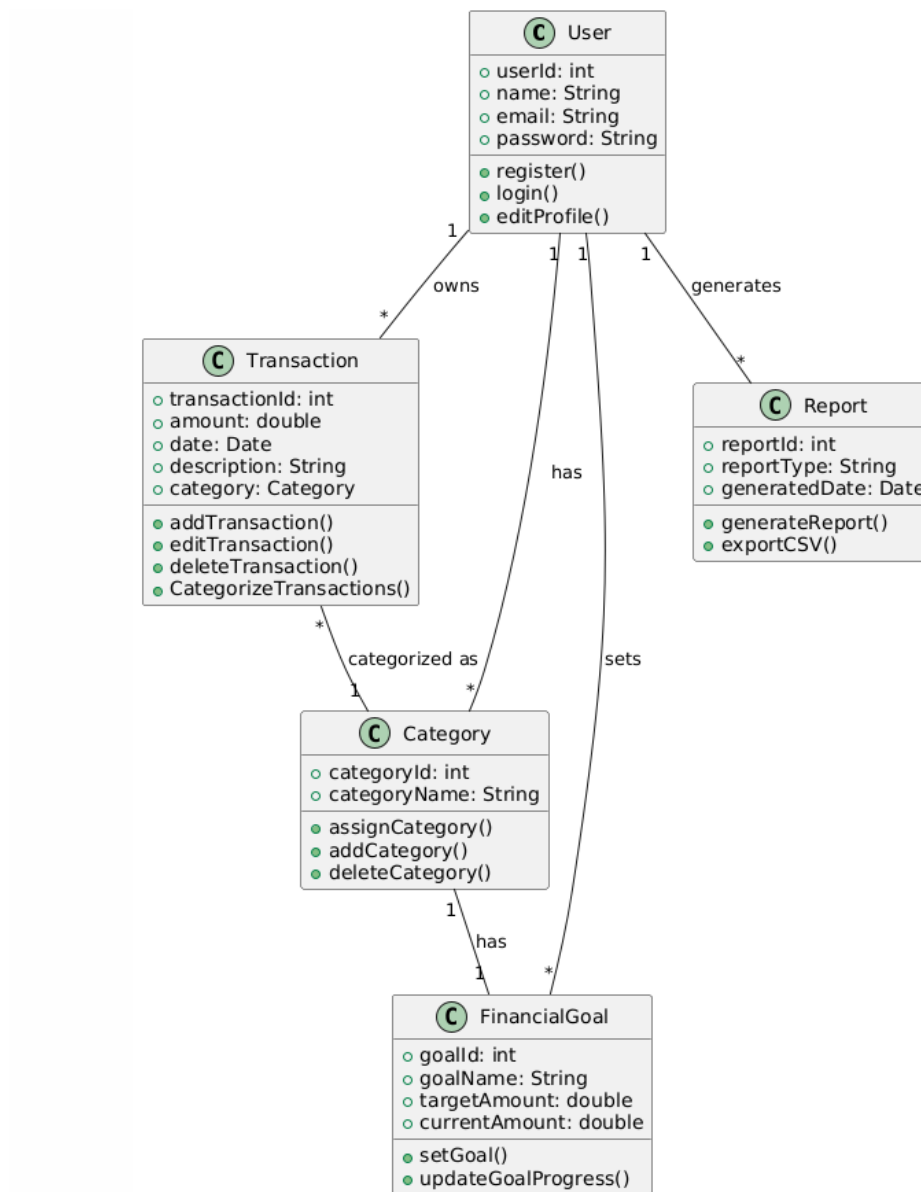


Figure 2.2: System Class Diagram

2.3 Pilotage avec SCRUM

2.3.1 Teams and Roles

Same team structure as presented in Chapter 1.

2.3.2 Product Backlog

ID	User Story	Priority
US1	As a user, I want to sign up with an email and password.	High

US2	As a user, I want to log in and log out.	High
US3	As a user, I want to edit my profile (name, password, photo).	Medium
US4	As a user, I want to add a transaction with a category, amount, and date.	High
US5	As a user, I want to edit or delete an existing transaction.	Medium
US6	As a user, I want to categorize my transactions (rent, groceries, salary, etc.).	High
US7	As a user, I want to see a summary of my income and expenses on my dashboard.	High
US8	As a user, I want to see graphs illustrating my expenses by category.	Medium
US9	As a user, I want to track my balance over time.	Medium
US10	As a user, I want to set a savings goal.	High
US11	As a user, I want to receive a notification when I reach a certain percentage of my goal.	Medium
US12	As a user, I want to generate a monthly financial report.	Medium
US13	As a user, I want to export my transactions as a CSV file.	Low
US14	As a user, I want my data to be securely encrypted.	High
US15	As a user, I want to send email to the support team.	Medium
US16	As a user, I want to switch to dark mode.	Low
US17	As a user, I want to use multiple currencies.	Medium
US18	As a user, I want to search for support options.	High

2.4 Sprint Plan - Spendora

2.4.1 Sprint 1: Authentication and User Management (2 weeks)

ID	User Story	Priority
US1	Sign up with an email and password.	High
US2	Log in and log out.	High
US3	Edit profile (name, password, photo).	Medium
US15	Send email to the support team.	Medium

Table 2.2: Sprint 1 Backlog

2.4.2 Sprint 2: Financial Transactions Management (2 weeks)

ID	User Story	Priority
US4	Add a transaction with a category, amount, and date.	High
US5	Edit or delete an existing transaction.	Medium
US6	Categorize transactions (rent, groceries, salary, etc.).	High

Table 2.3: Sprint 2 Backlog

2.4.3 Sprint 3: Dashboard and Visualization (2 weeks)

ID	User Story	Priority
US7	View a summary of income and expenses on the dashboard.	High
US8	View expense graphs by category.	Medium
US9	Track balance over time.	Medium

Table 2.4: Sprint 3 Backlog

2.4.4 Sprint 4: Financial Goals and Notifications (1 week)

ID	User Story	Priority
US10	Set a savings goal.	High
US11	Receive a notification when a goal is reached.	Medium

Table 2.5: Sprint 4 Backlog

2.4.5 Sprint 5: Reports and Exporting (1 week)

ID	User Story	Priority
US12	Generate a monthly financial report.	Medium
US13	Export transactions as a CSV file.	Low

Table 2.6: Sprint 5 Backlog

2.4.6 Sprint 6: Security and Optimization (1 week)

ID	User Story	Priority
US14	Secure data with encryption.	High
US16	Switch to dark mode.	Low
US17	Use multiple currencies.	Medium
US18	Search for support options.	High

Table 2.7: Sprint 6 Backlog

2.5 Configuration and Development Environment

2.5.1 Development Architecture



Figure 2.3: Development Environment Architecture

2.5.2 Development Tools

2.5.2.1 Database Tools

- **MySQL:** A relational database management system (RDBMS). One of the most widely used database systems in the world.
- **phpMyAdmin:** A web application for MySQL and MariaDB management.

2.5.2.2 Mobile Development Tools

- **Flutter:** A user interface development kit created by Google.
- **Visual Studio Code:** An extensible code editor developed by Microsoft.

2.5.2.3 Design Tools

- **Draw.io**: Diagram creation software.
- **Figma**: Collaborative interface design tool.
- **Canva**: Graphic design platform.
- **Photoshop**: Professional image editor.

2.5.3 Technologies and Languages Used

2.5.3.1 Frontend Technologies

- **Dart**: Client-optimized programming language.
- **Flutter SDK**: Cross-platform user interface framework.

2.5.3.2 Backend Technologies

- **Python**: Main programming language
- **Flask**: Lightweight web framework
- **PyMySQL**: MySQL connection library
- **Matplotlib**: Data visualization library

2.6 Conclusion

This chapter has provided an in-depth examination of the technical specifications and design aspects of Spendora. Through careful analysis of user requirements and market needs, we've established a robust foundation for our application's architecture and functionality. The comprehensive sprint planning demonstrates our commitment to delivering features in a systematic and prioritized manner, ensuring that core functionalities are implemented first while maintaining flexibility for enhancement.

The technical stack selection has been deliberately chosen to ensure scalability, performance, and maintainability of the application. Our development environment configuration reflects modern best practices in software development, providing team members with the tools and frameworks necessary for efficient collaboration and code delivery.

Furthermore, the detailed user story mapping and sprint organization provide a clear path forward for development while maintaining agile principles. This approach will allow us to adapt to changing requirements while ensuring consistent progress toward our project goals. The next phases of the project will focus on implementing these specifications while maintaining high standards of code quality and user experience.

Chapter 3

« Study and Implementation of Sprint 1: Authentication and Profile Management »

3.1 Introduction

This chapter presents in detail the implementation of our project's first sprint, which focuses on user authentication and profile management. This phase is fundamental as it constitutes the security and access foundation of our Spendora application.

3.2 Sprint 1 Backlog

ID	User Story	Priority	Points
US1	Sign up with email and password	High	5
US2	Login and logout	High	3
US3	Edit profile (name, password, photo)	Medium	5
US4	Reset password	Medium	3

Table 3.8: Sprint 1 Backlog

3.3 Functional Specification

3.3.1 Use Case Diagram

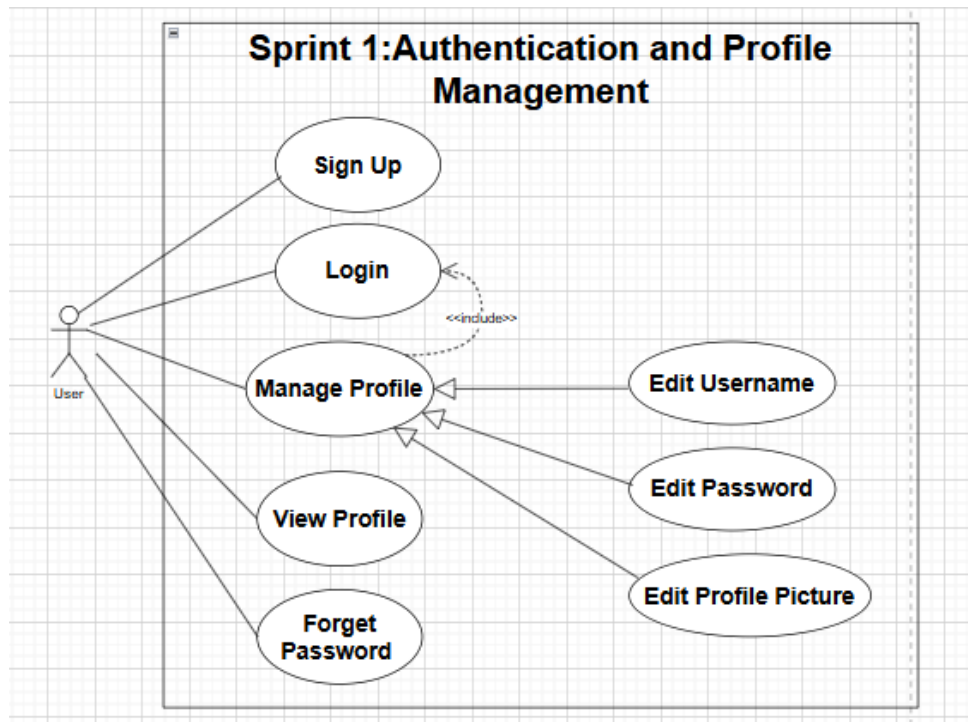


Figure 3.4: Sprint 1 Use Case Diagram

3.3.2 Textual Description of Use Cases

3.3.2.1 Use Case: Sign Up

- **Main Actor:** Unauthenticated User

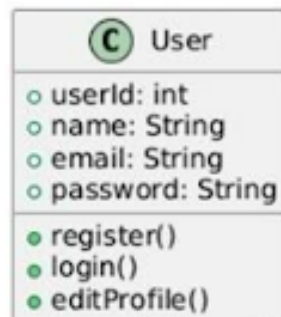
- **Precondition:** None
- **Main Scenario:**
 1. The user accesses the sign-up page
 2. The user enters their email and password
 3. The system verifies the validity of the data
 4. The system creates the user account
 5. The system sends a confirmation email
- **Postcondition:** Account created and confirmation email sent

3.3.2.2 Use Case: Login

- **Main Actor:** Unauthenticated User
- **Precondition:** Have a valid account
- **Main Scenario:**
 1. The user enters their credentials
 2. The system verifies the credentials
 3. The system grants access
- **Postcondition:** User logged in

3.4 Design

3.4.1 Class Diagram



3.4.2 Detailed Sequence Diagrams

3.4.2.1 Authentication Sequence

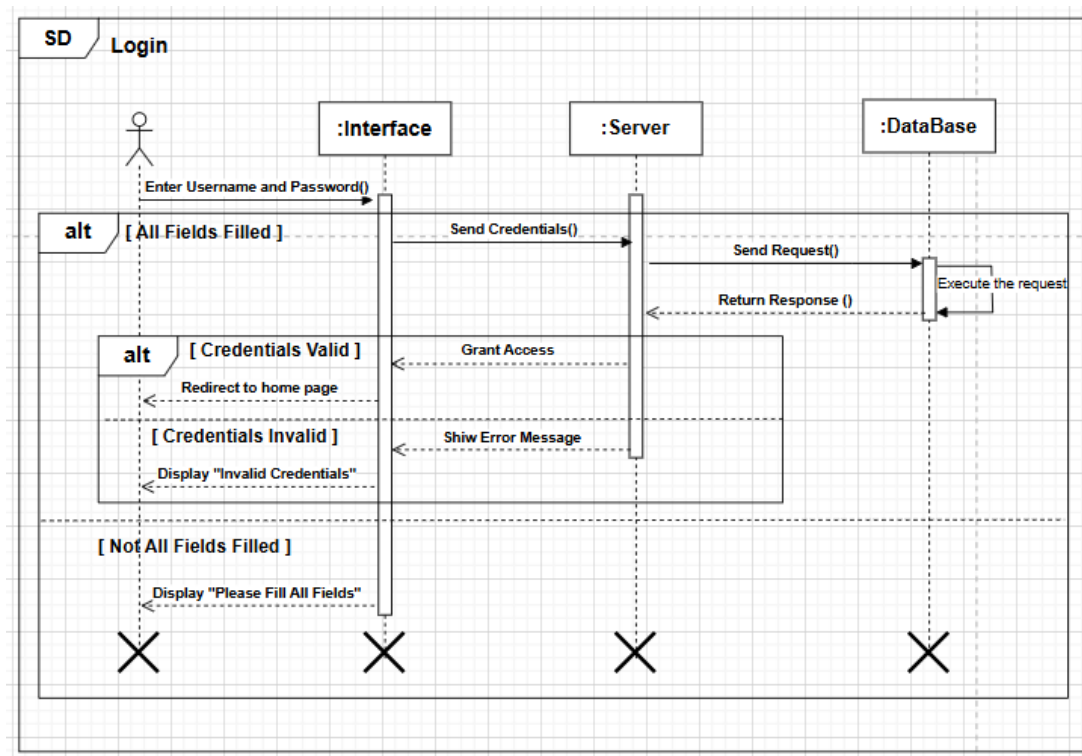


Figure 3.6: Sequence Diagram - Login Process

3.4.2.2 Sign-Up Sequence

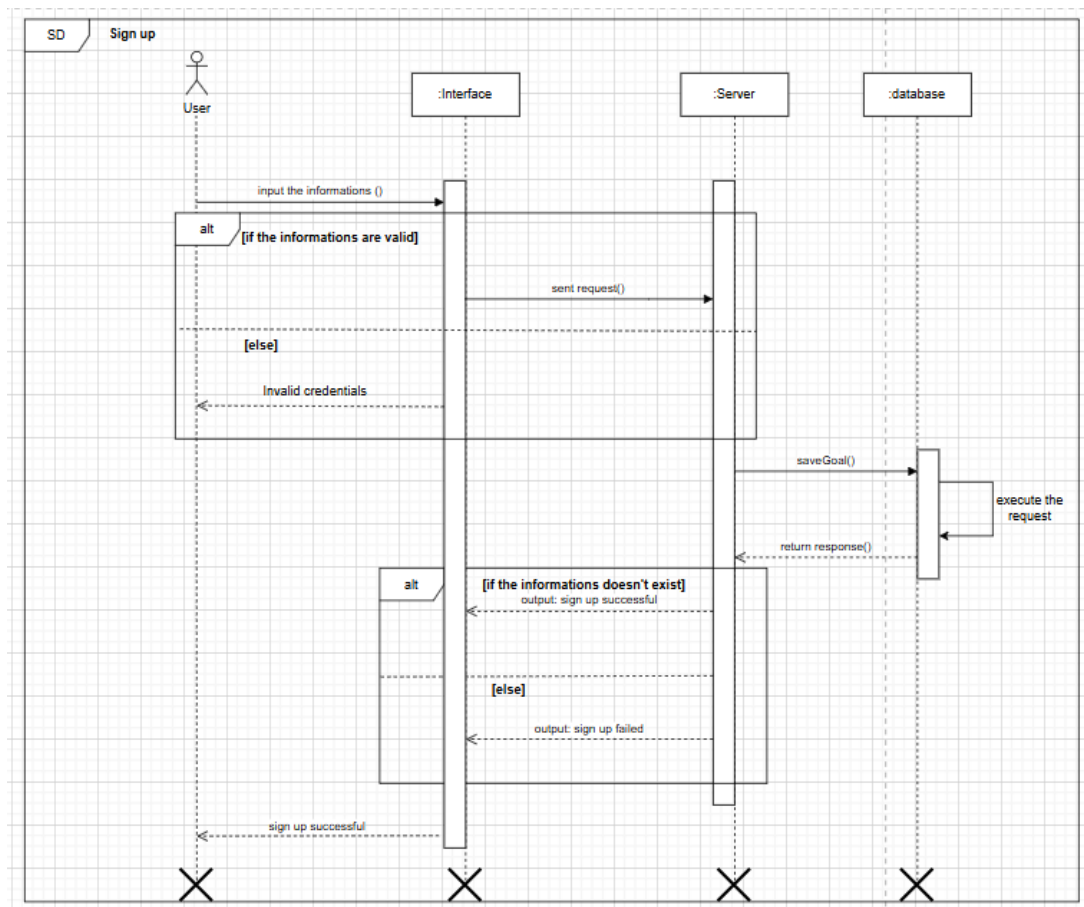


Figure 3.7: Sequence Diagram - Sign-Up Process

3.4.2.3 Password Reset Sequence

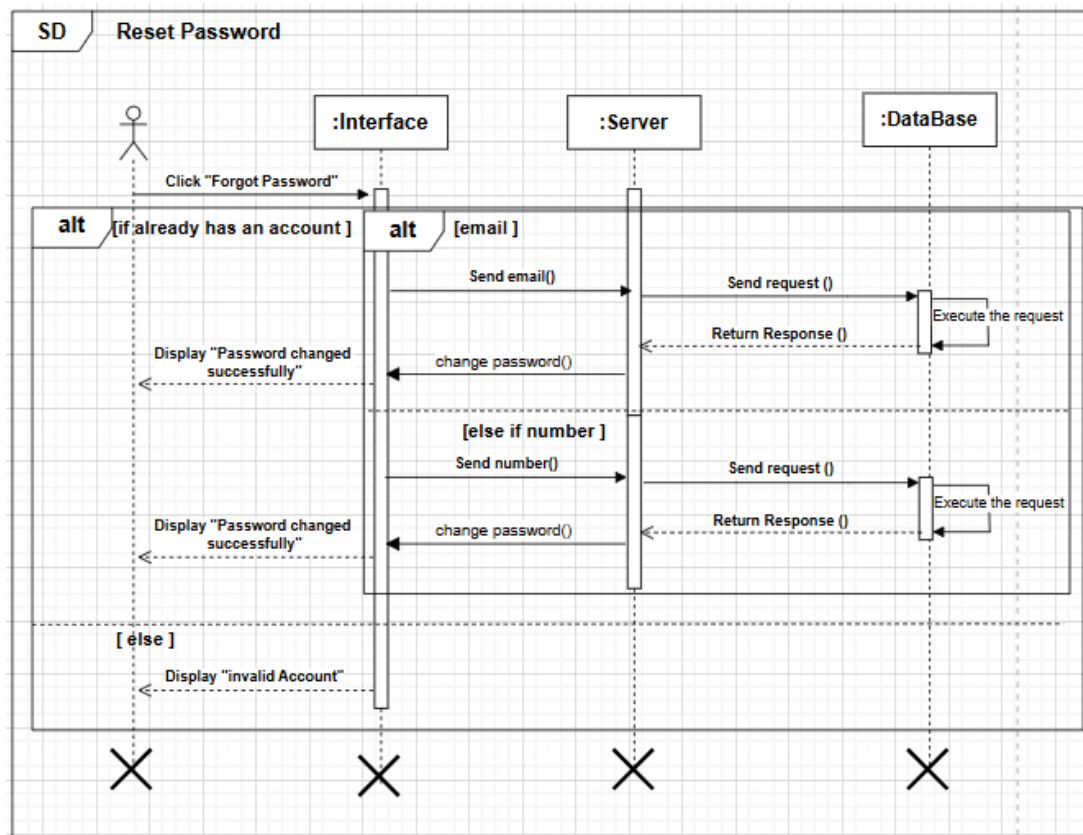


Figure 3.8: Sequence Diagram - Password Reset

3.4.2.4 Interaction Descriptions

Login Process:

1. The user enters their credentials
2. The client sends the credentials to the server
3. The server verifies the credentials in the database
4. On success, the server generates a JWT token
5. The token is sent back to the client
6. The client stores the token and redirects to the dashboard

Sign-Up Process:

1. The user fills out the sign-up form
2. The client validates the data format
3. The data is sent to the server
4. The server checks the email's uniqueness

5. The server hashes the password
6. The data is saved in the database
7. A confirmation email is sent

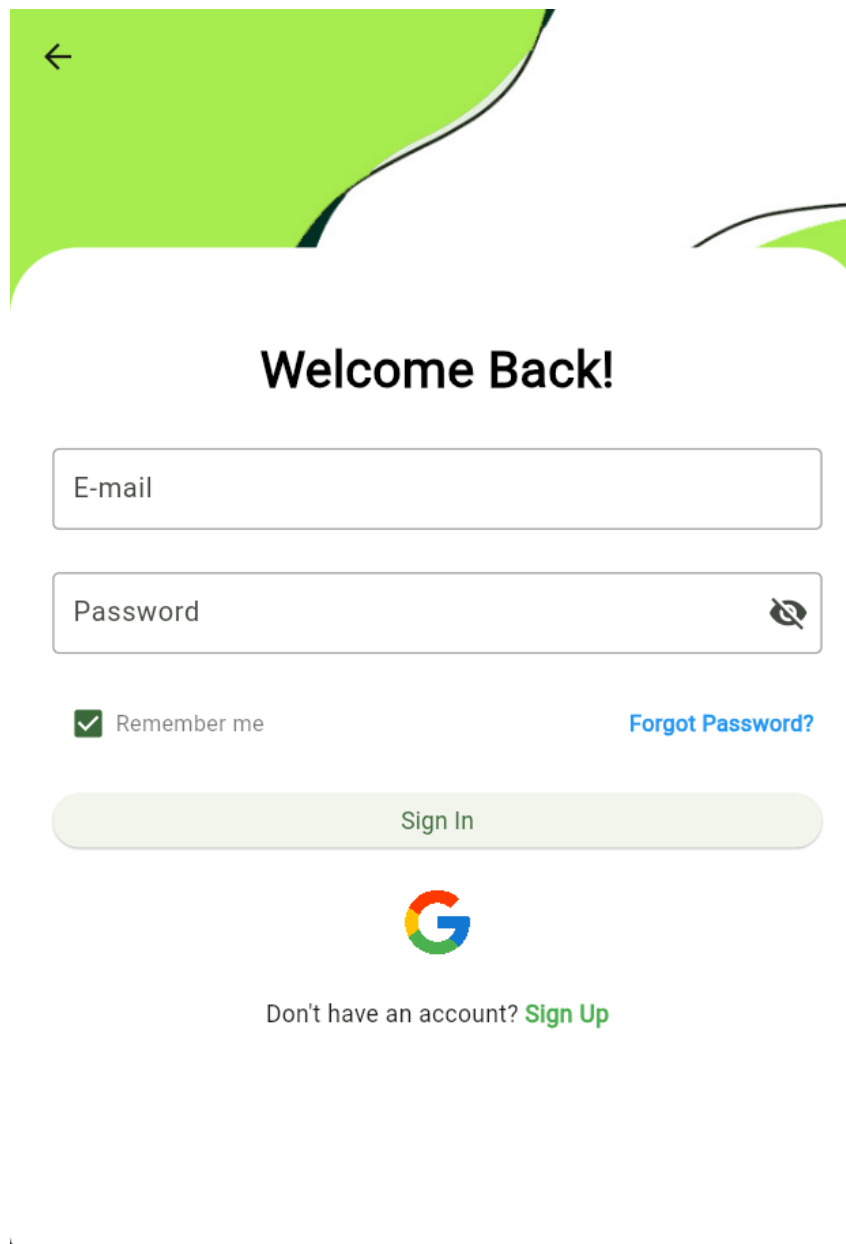
Password Reset:

1. The user requests a password reset
2. The server generates a unique token
3. An email with the reset link is sent
4. The user sets a new password
5. The server verifies the token and updates the password

3.5 Implementation

This section presents the user interfaces developed during this sprint.

3.5.1 Authentication Interface



The image shows a login interface with a green header bar containing a back arrow. Below the header, the text "Welcome Back!" is centered. There are two input fields: "E-mail" and "Password". The "Password" field has a toggle icon (an eye with a slash) on the right. Below the "E-mail" field, there is a checked checkbox labeled "Remember me" and a link "Forgot Password?". A large green button labeled "Sign In" is centered below the "Remember me" checkbox. Below the "Sign In" button is the Google "G" logo. At the bottom, there is a link "Don't have an account? Sign Up".

←

Welcome Back!

E-mail

Password 

☒ Remember me [Forgot Password?](#)

Sign In



Don't have an account? [Sign Up](#)

Figure 3.9: Login Interface

3.5.2 Profile Management Interface

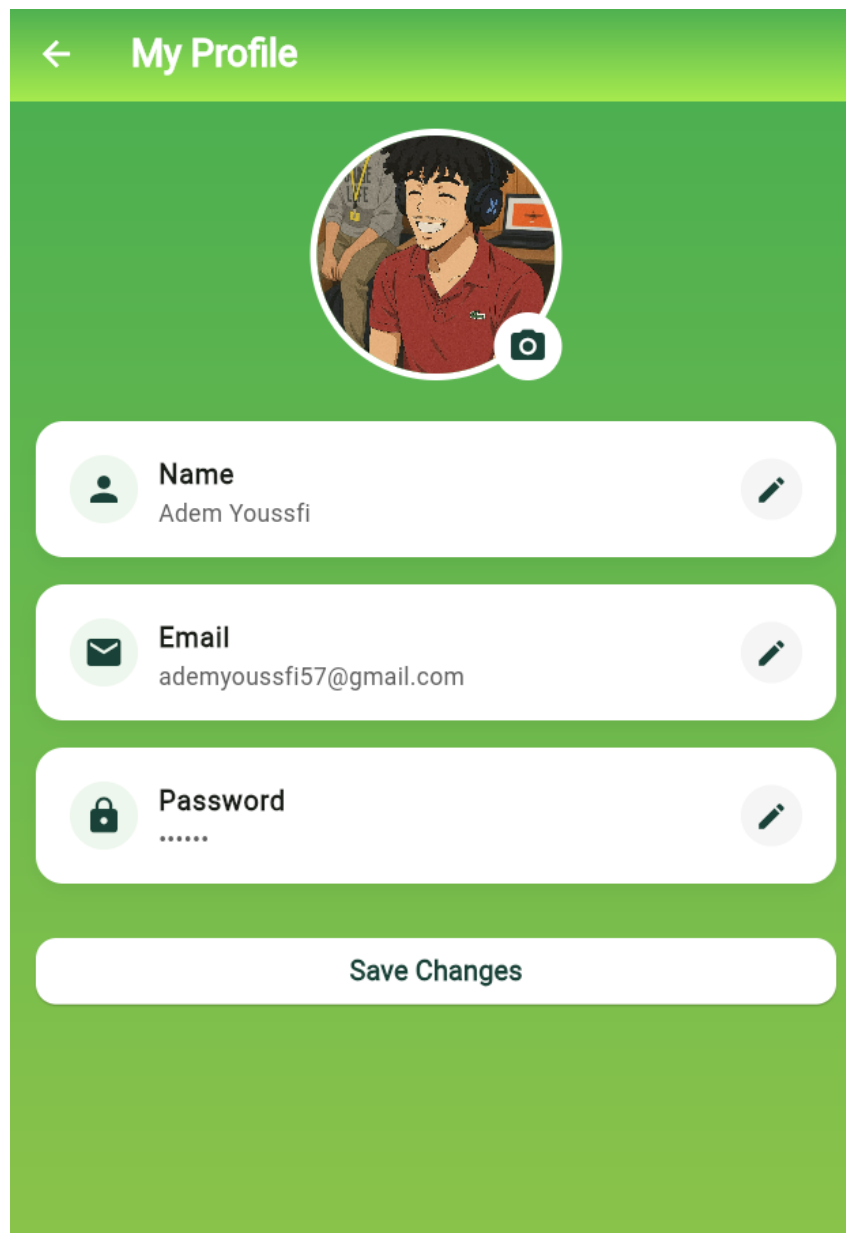
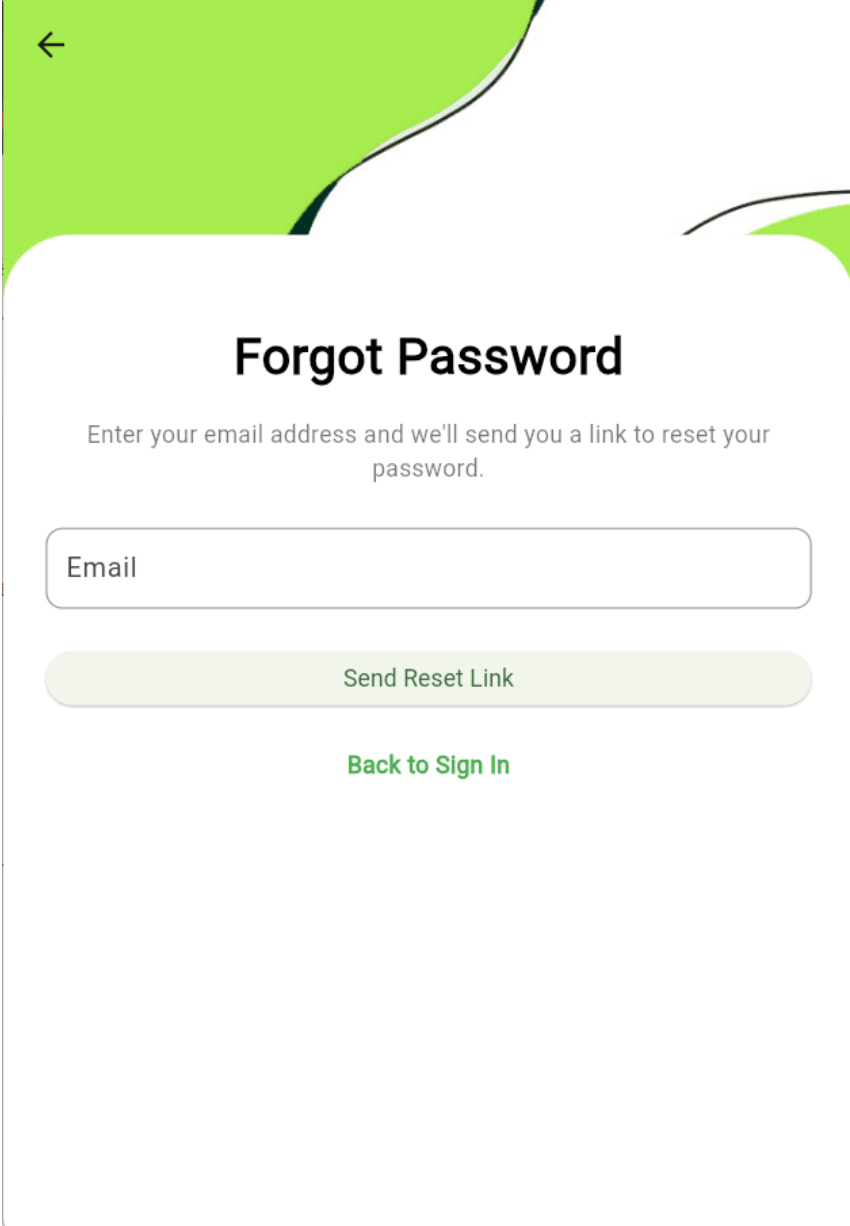


Figure 3.10: Profile Management Interface

3.5.3 Forgot Password Interface

The image shows a mobile application interface for a password reset. At the top left, there is a green header bar with a white back arrow icon. Below the header, the main content area has a white background with rounded corners. The title "Forgot Password" is centered in a bold, black font. Below the title, a subtitle in a smaller, gray font reads: "Enter your email address and we'll send you a link to reset your password." There is a white rectangular input field with rounded corners and a thin gray border, containing the placeholder text "Email". Below the input field is a light green button with rounded corners and a thin gray border, containing the text "Send Reset Link". At the bottom, there is a green link text "Back to Sign In" in a medium-sized font.

←

Forgot Password

Enter your email address and we'll send you a link to reset your password.

Send Reset Link

[Back to Sign In](#)

Figure 3.11: Password Reset Interface

3.5.4 Sign-Up Request Interface

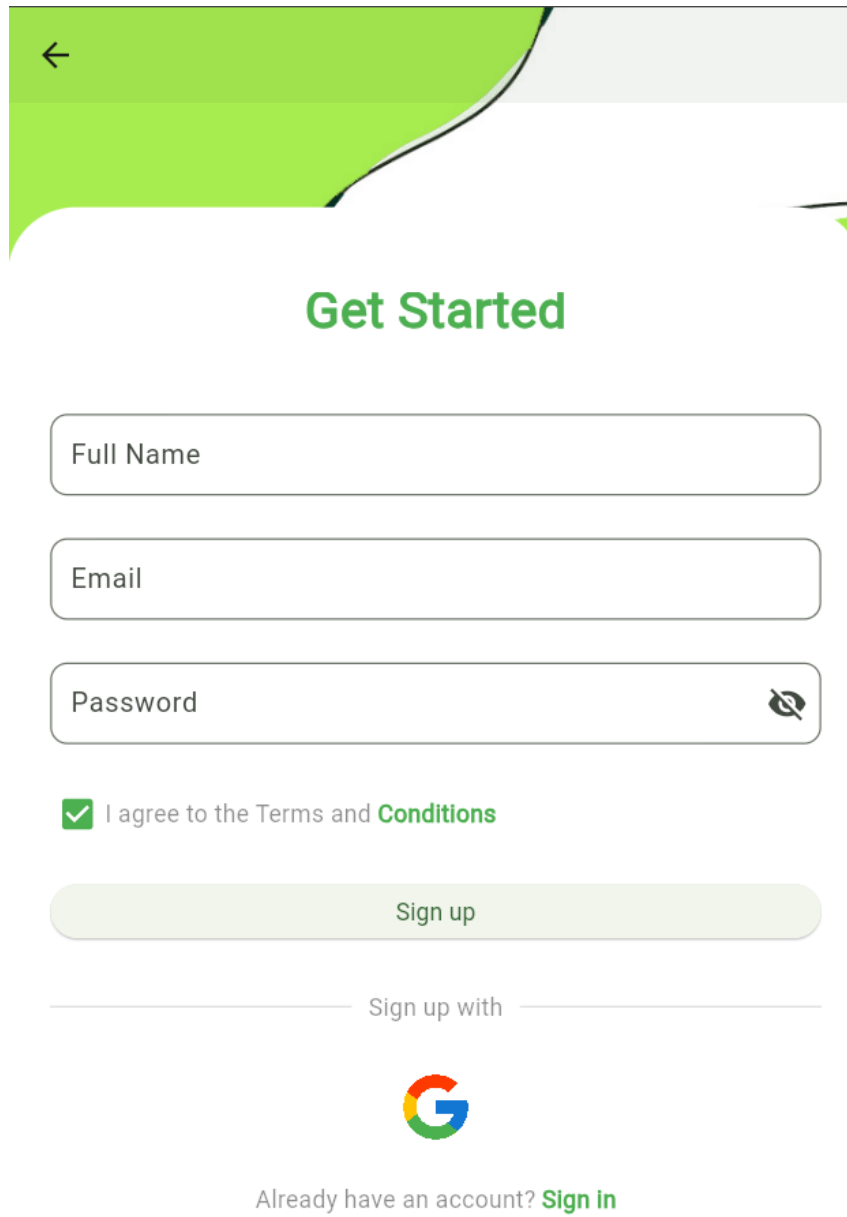
The image shows a mobile application sign-up screen. At the top, there is a green header bar with a white back arrow on the left. Below the header, the text "Get Started" is displayed in a bold, green font. The form consists of three rounded rectangular input fields: "Full Name", "Email", and "Password". The "Password" field has a small eye icon on the right to toggle visibility. Below the fields, there is a green checkmark icon followed by the text "I agree to the Terms and Conditions". A large, light green "Sign up" button is positioned below the checkbox. At the bottom, there is a horizontal line with the text "Sign up with" in the center, followed by the Google logo. Below the logo, the text "Already have an account? Sign in" is displayed, with "Sign in" in green.

Figure 3.12: Sign-Up Interface

3.6 Conclusion

This first sprint established the essential foundations of our Spendora application. The implementation of authentication and profile management functionalities provides a solid base for future developments. The user interfaces were designed to be intuitive and accessible, while the backend architecture ensures system security and reliability.

The objectives set for this sprint were successfully achieved, including:

- Establishing a robust authentication system
- Developing ergonomic user interfaces

- Implementing profile management
- Securing user data

Initial user feedback is positive, and the developed functionalities will provide a solid foundation for implementing future financial management features in subsequent sprints.

Chapter 4

« Study and Implementation of Sprint
2:
Financial Transaction Management »

4.1 Introduction

This chapter presents in detail the implementation of our project's second sprint, which focuses on financial transaction management. This phase is crucial as it represents the core functionality of our Spendora application.

4.2 Sprint 2 Backlog

ID	User Story	Priority	Points
US5	Add a transaction with category, amount and date	High	5
US6	Edit or delete an existing transaction	Medium	3
US7	Categorize transactions	High	5
US8	View transaction history	Medium	3

Table 4.9: Sprint 2 Backlog

4.3 Functional Specification

4.3.1 Use Case Diagram

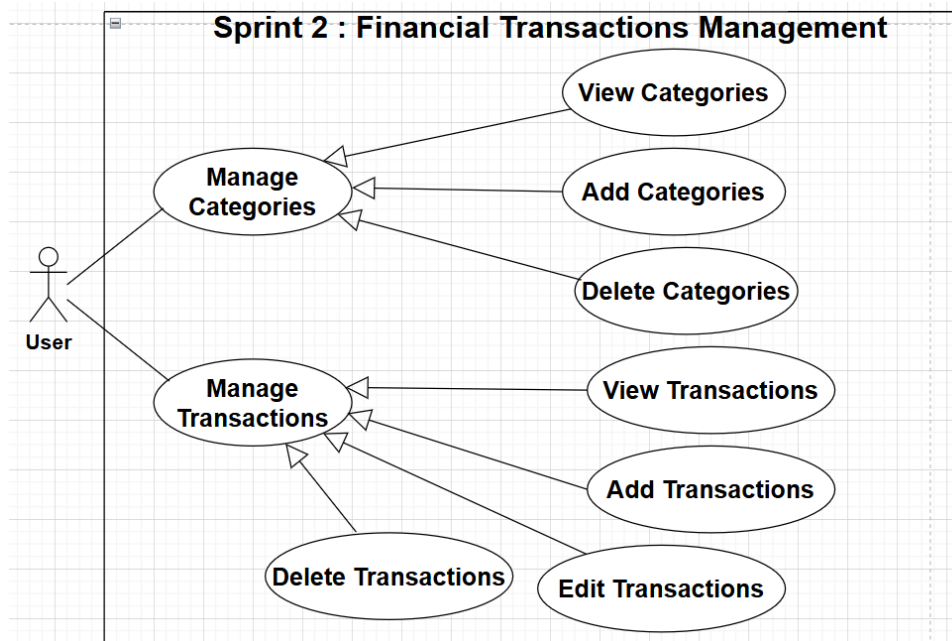


Figure 4.13: Use Case Diagram for Sprint 2

4.3.2 Textual Description of Use Cases

4.3.2.1 Use Case: Add Transaction

- **Main Actor:** Authenticated User
- **Precondition:** Be logged into the system

- **Main Scenario:**

1. User accesses the transaction addition form
2. User enters transaction details
3. System validates the data
4. System records the transaction
5. System updates account balance

- **Postcondition:** Transaction recorded and balance updated

4.3.2.2 Use Case: Manage Categories

- **Main Actor:** Authenticated User

- **Precondition:** Be logged into the system

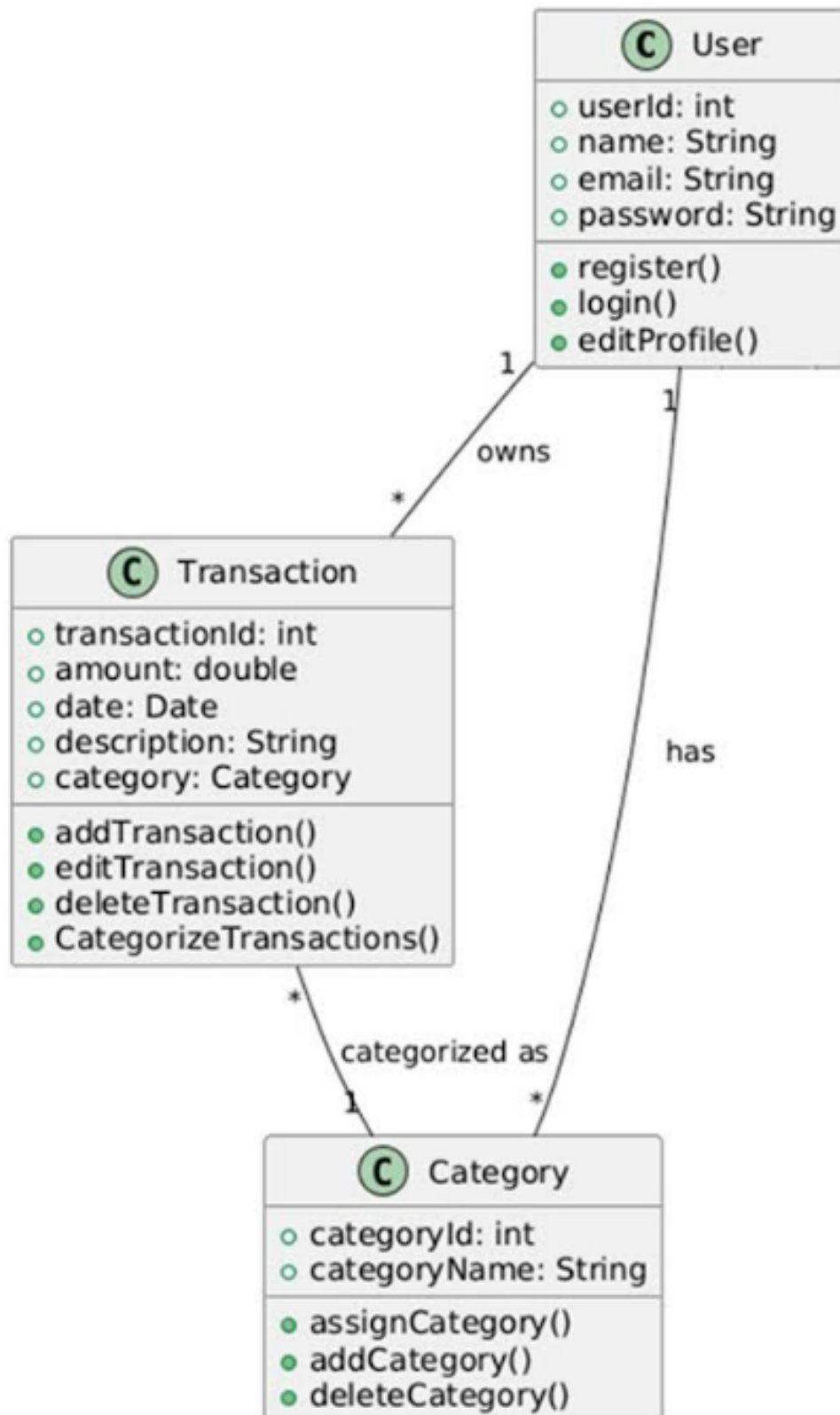
- **Main Scenario:**

1. User accesses category management
2. User can create/edit/delete categories
3. System updates the category list

- **Postcondition:** Categories updated

4.4 Design

4.4.1 Class Diagram



4.4.2 Detailed Sequence Diagrams

4.4.2.1 Transaction Addition Sequence

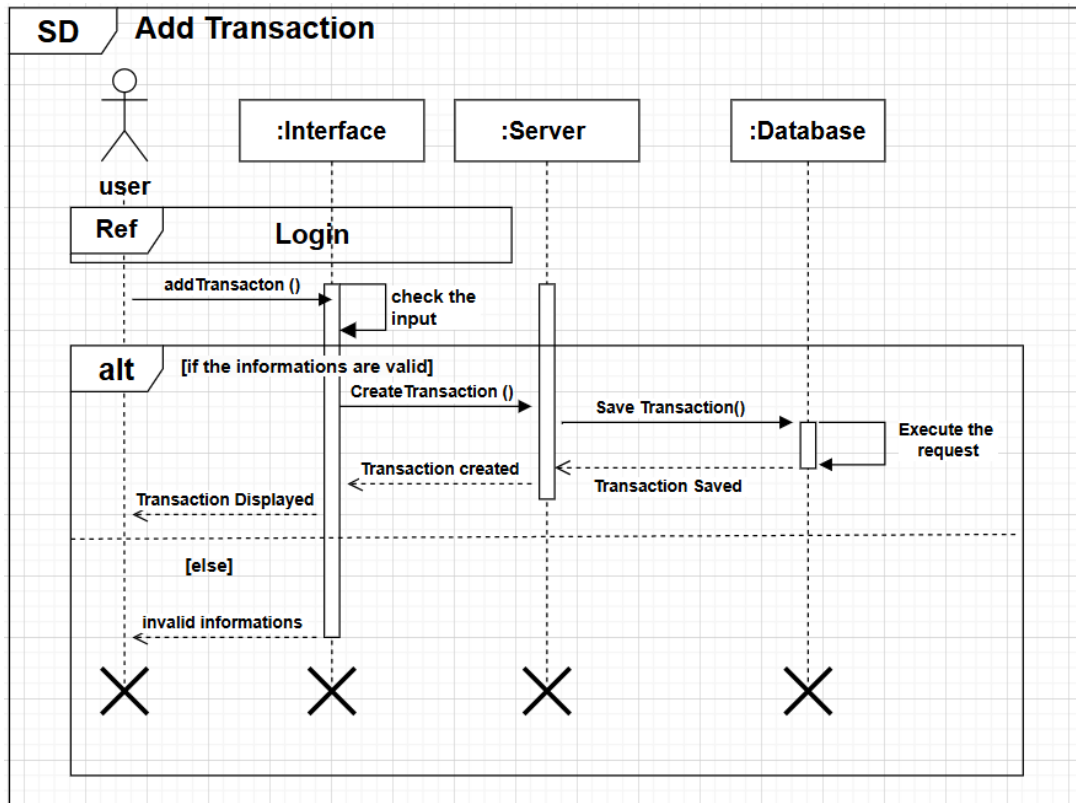


Figure 4.15: Sequence Diagram - Transaction Addition

4.4.2.2 Interaction Description

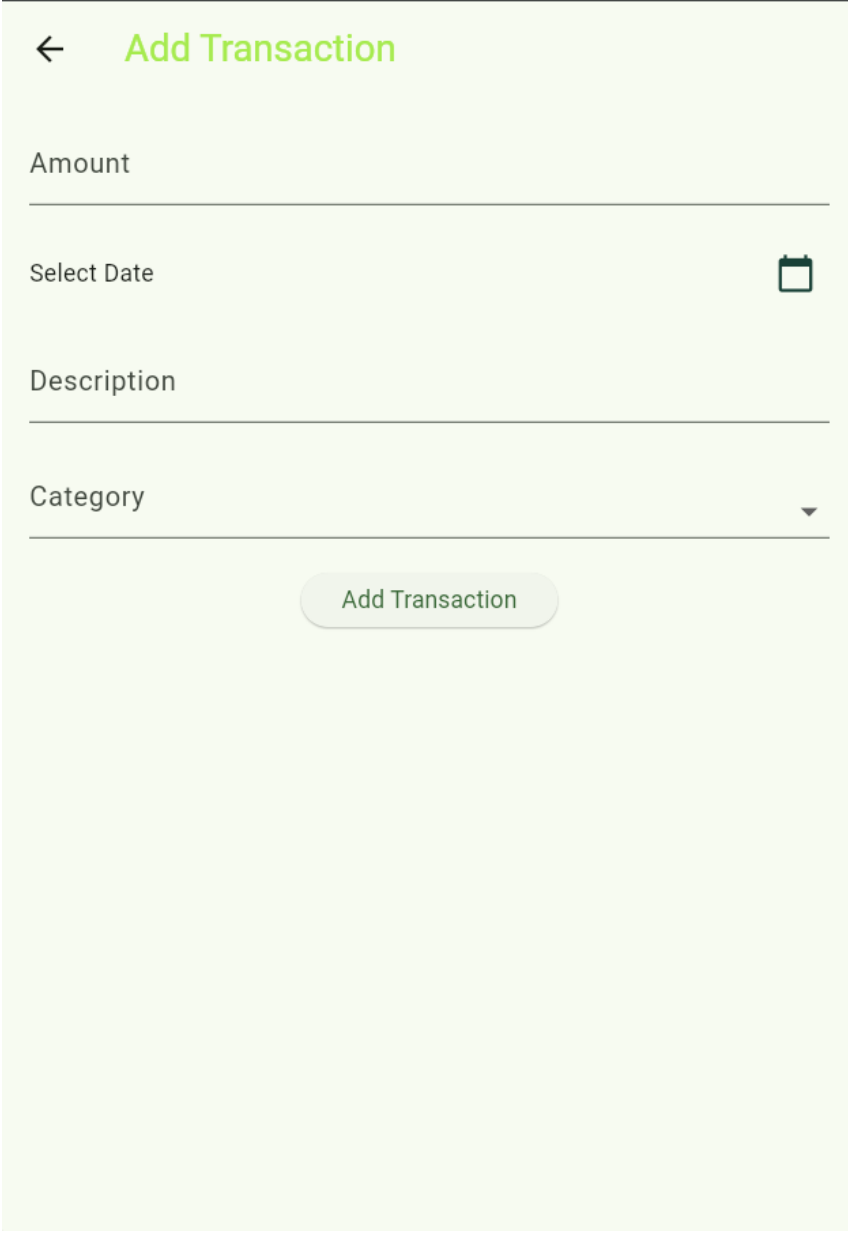
Transaction Addition Process:

1. User fills out the transaction form
2. Client validates the entered data
3. Data is sent to the server
4. Server checks the validity of the data
5. Transaction is recorded in the database
6. Account balance is updated
7. Confirmation is sent to the client

4.5 Implementation

This section presents the user interfaces developed during this sprint.

4.5.1 Transaction Addition Interface



The image shows a mobile application interface for adding a transaction. It features a light green background. At the top, there is a header bar with a back arrow icon on the left and the text "Add Transaction" in green. Below the header, there are four input fields, each with a label and a horizontal line for text entry: "Amount", "Select Date" (which includes a calendar icon on the right), "Description", and "Category" (which includes a downward arrow icon on the right). At the bottom of the form, there is a rounded rectangular button with the text "Add Transaction".

Figure 4.16: Transaction Addition Interface

4.5.2 Transaction List Interface

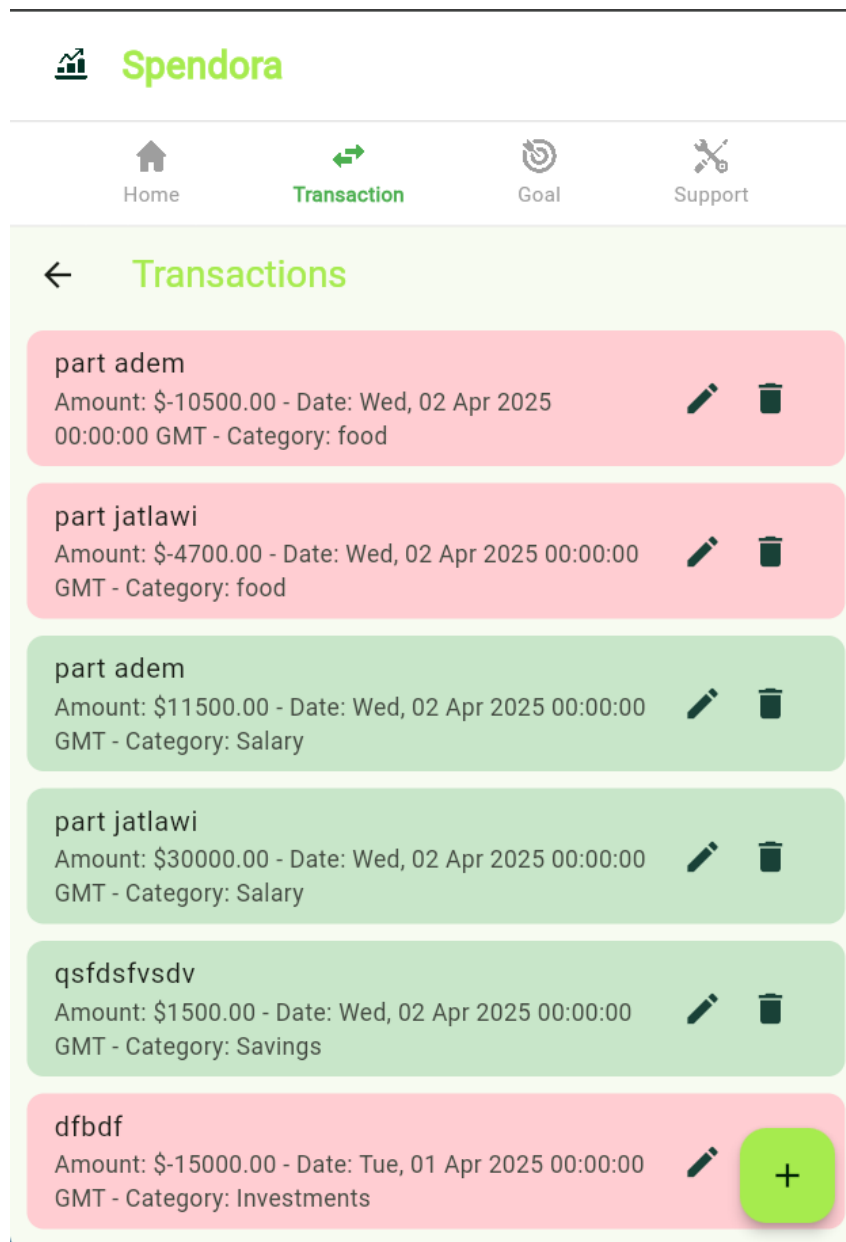


Figure 4.17: Transaction List Interface

4.5.3 Category Management Interface

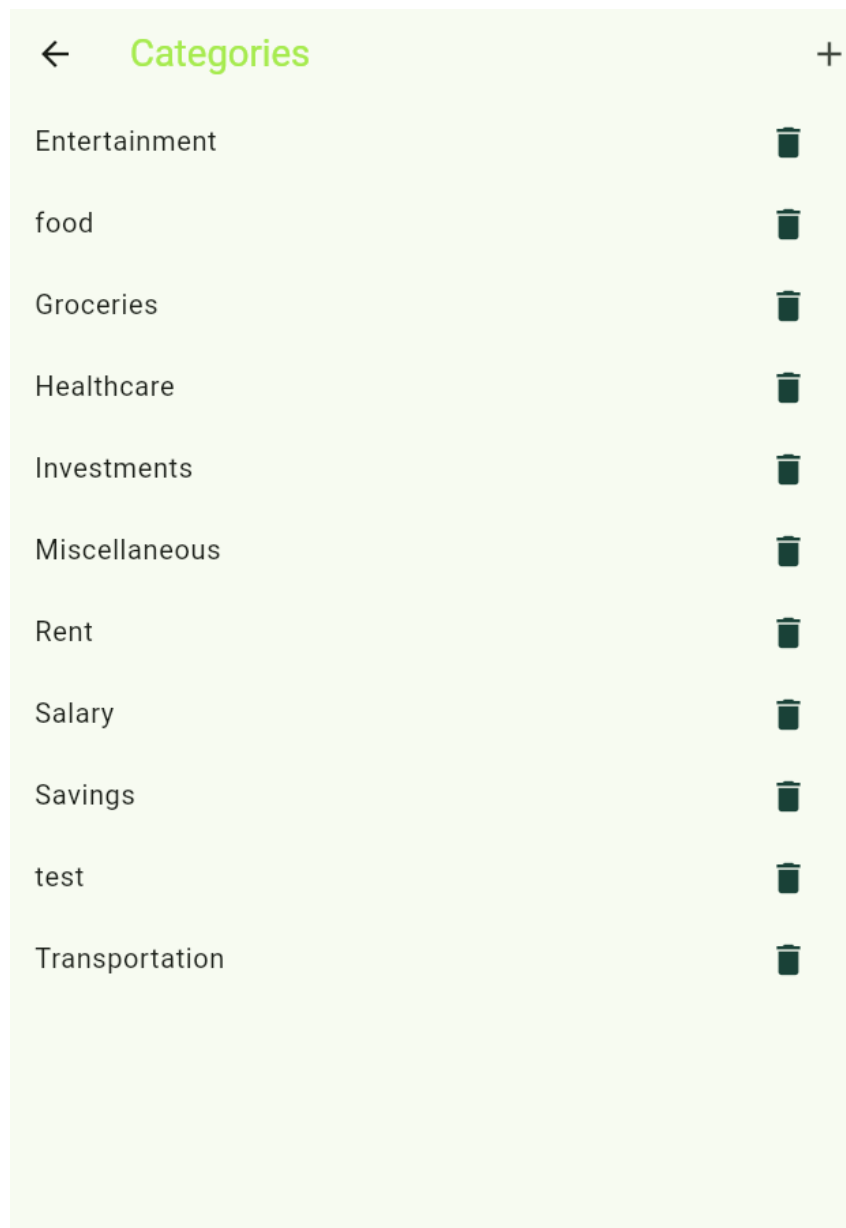


Figure 4.18: Category Management Interface

4.6 Conclusion

This second sprint has implemented the essential financial transaction management features of Spendora. The architecture put in place enables efficient and secure transaction management while offering an intuitive and flexible user experience.

The main objectives were successfully achieved:

- Robust implementation of the transaction management system
- Implementation of a flexible categorization system
- Development of intuitive user interfaces

- Validation and security of financial data

The features developed in this sprint constitute the core of our application and will serve as a foundation for the analytical features to be developed in subsequent sprints.

Chapter 5

« Study and Implementation of Sprint 3: Dashboard and Visualization »

5.1 Introduction

This chapter presents the implementation of our project's third sprint, focused on developing the dashboard and financial data visualization features. This phase is essential to enable users to understand and analyze their financial habits.

5.2 Sprint 3 Backlog

ID	User Story	Priority	Points
US9	View income and expense summary on dashboard	High	5
US10	Visualize expenses by category in graphs	High	8
US11	Track balance evolution over time	Medium	5
US12	Generate monthly reports	Medium	5

Table 5.10: Sprint 3 Backlog

5.3 Functional Specification

5.3.1 Use Case Diagram

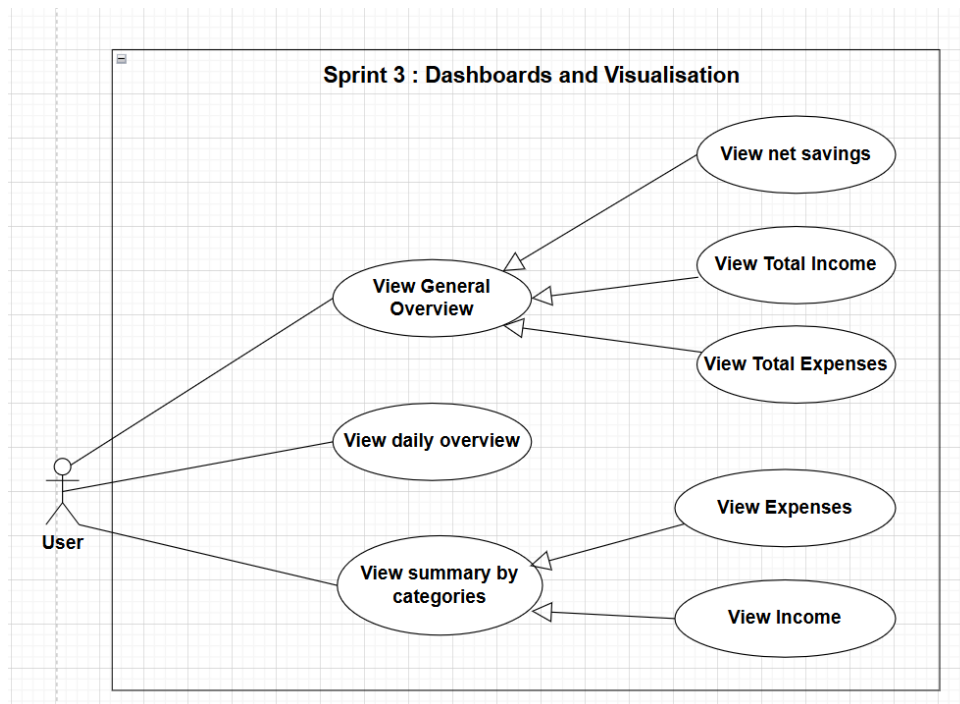


Figure 5.19: Sprint 3 Use Case Diagram

5.3.2 Use Case Description

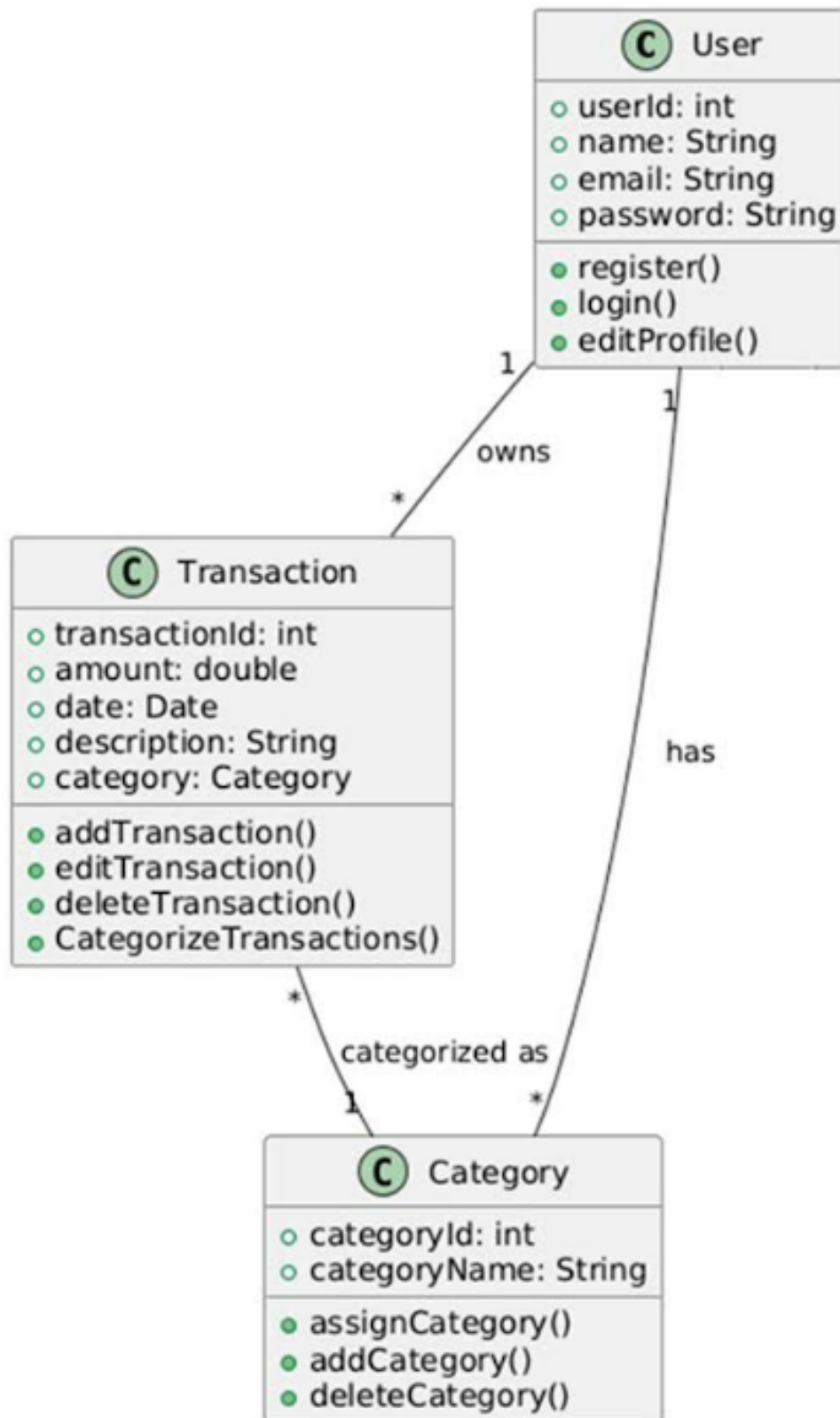
5.3.2.1 Use Case: View Dashboard

- **Main Actor:** Authenticated User

- **Precondition:** Be logged in and have recorded transactions
- **Main Scenario:**
 1. User accesses the dashboard
 2. System calculates financial statistics
 3. System generates graphs
 4. System displays consolidated information
- **Postcondition:** Financial data visualized

5.4 Design

5.4.1 Class Diagram



5.4.2 Detailed Sequence Diagrams

5.4.2.1 Dashboard Generation Sequence

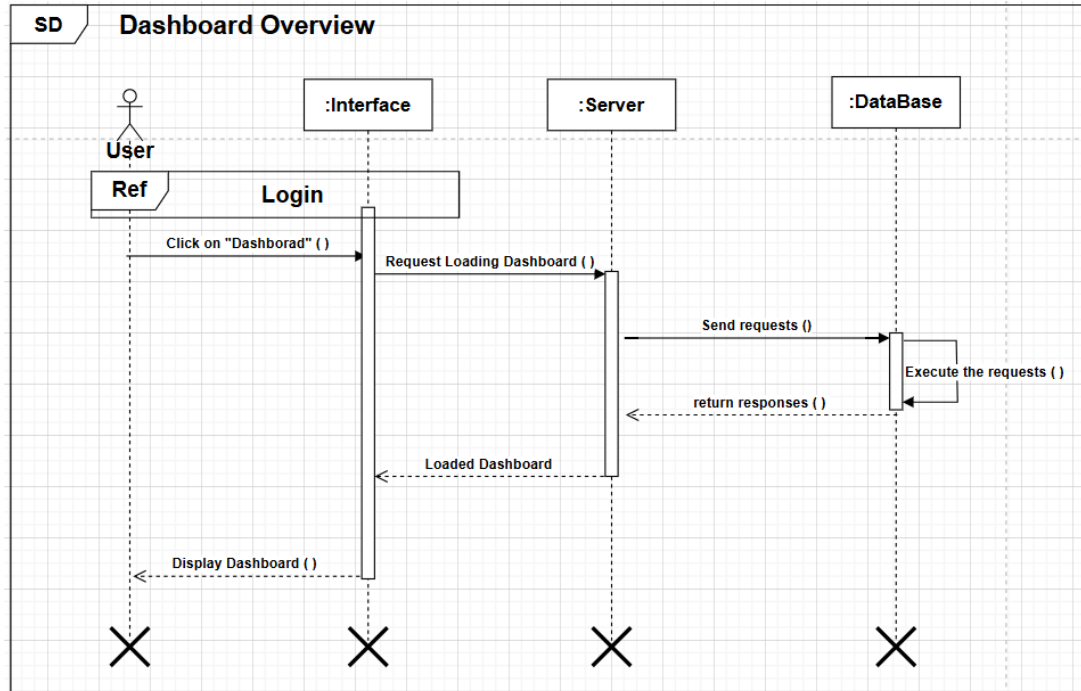


Figure 5.21: Sequence Diagram - Dashboard Generation

5.4.2.2 Visualization Generation Process:

1. Client requests dashboard data
2. Server retrieves transactions
3. Server calculates statistics
4. Server generates graph data
5. Client receives data
6. Client generates visualizations with Flutter

5.5 Implementation

This section presents the visualization interfaces developed.

5.5.1 Dashboard Interface

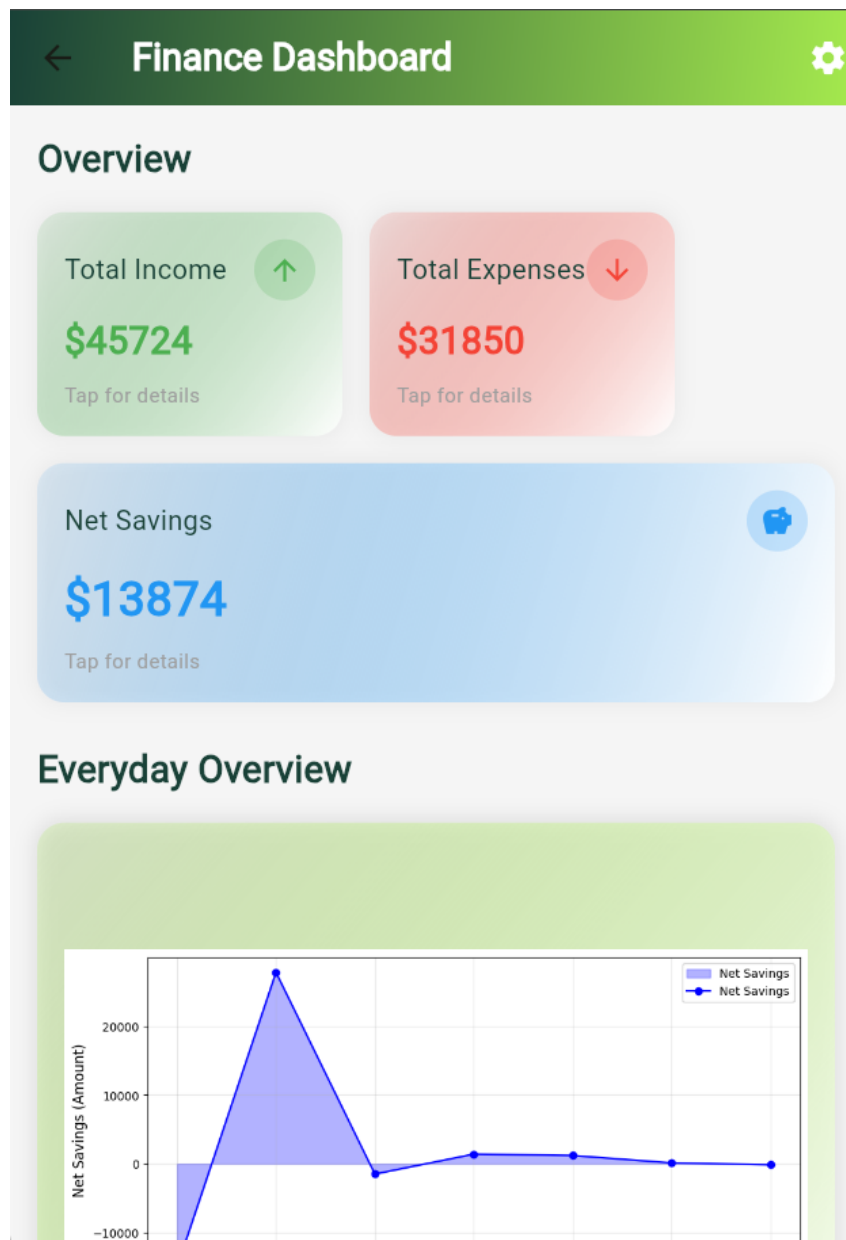


Figure 5.22: Main Dashboard Interface

5.5.2 Balance Evolution Chart

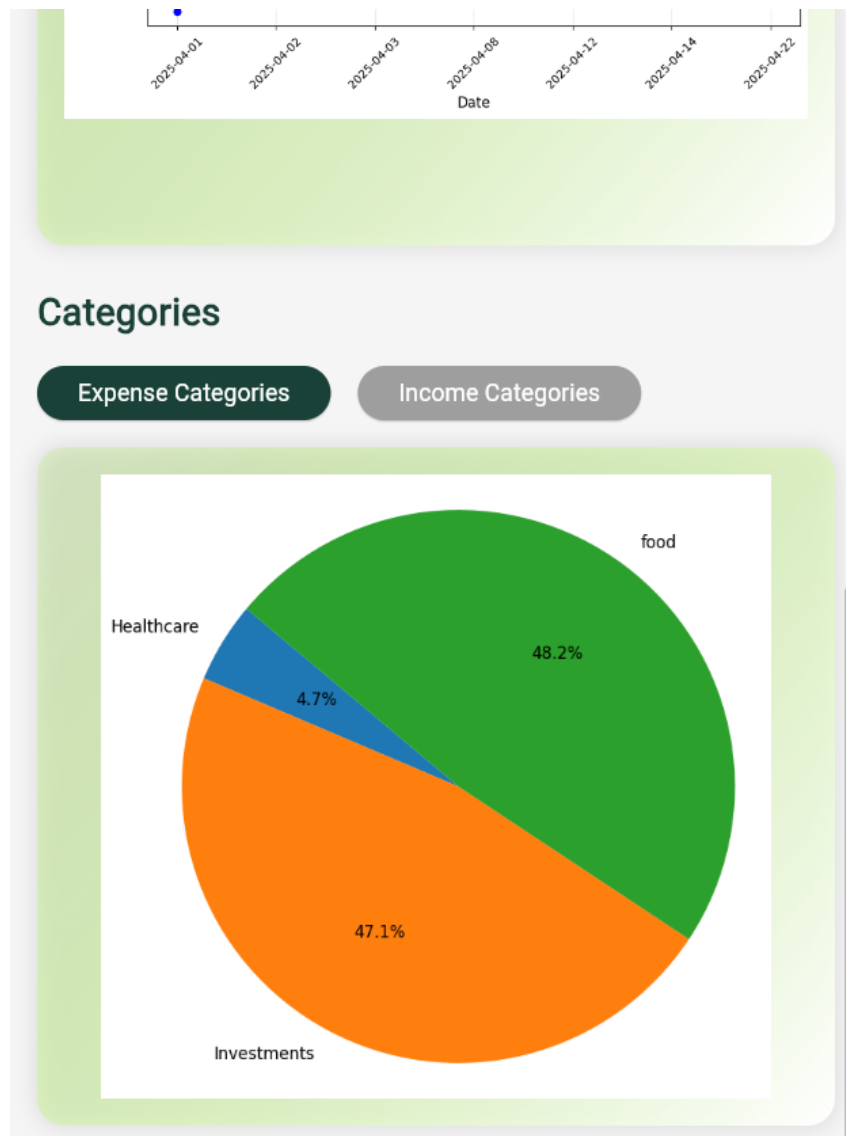


Figure 5.23: Balance Evolution Chart

5.6 Conclusion

This third sprint has significantly enhanced the user experience of Spendora by adding essential visualization features. The following objectives were achieved:

- Implementation of an intuitive and informative dashboard
- Creation of relevant graphical visualizations
- Implementation of financial analysis tools
- Optimization of graph generation performance

User feedback on the visualization features has been very positive, particularly highlighting the clarity of the graphs and the ease of interpreting financial data. These developments represent an important step in our goal of making financial management more accessible and understandable for all users.

Chapter 6

« Study and Implementation of Sprint
4:
Financial Goals and Notifications »

6.1 Introduction

This chapter presents the implementation of the fourth sprint, which focuses on financial goals management and notification system. This phase is important to help users achieve their savings goals and stay informed of their progress.

6.2 Sprint 4 Backlog

ID	User Story	Priority	Points
US13	Define a savings goal	High	5
US14	Track progress towards goal	High	5
US15	Receive progress notifications	Medium	3
US16	Manage notification settings	Low	2

Table 6.11: Sprint 4 Backlog

6.3 Functional Specification

6.3.1 Use Case Diagram

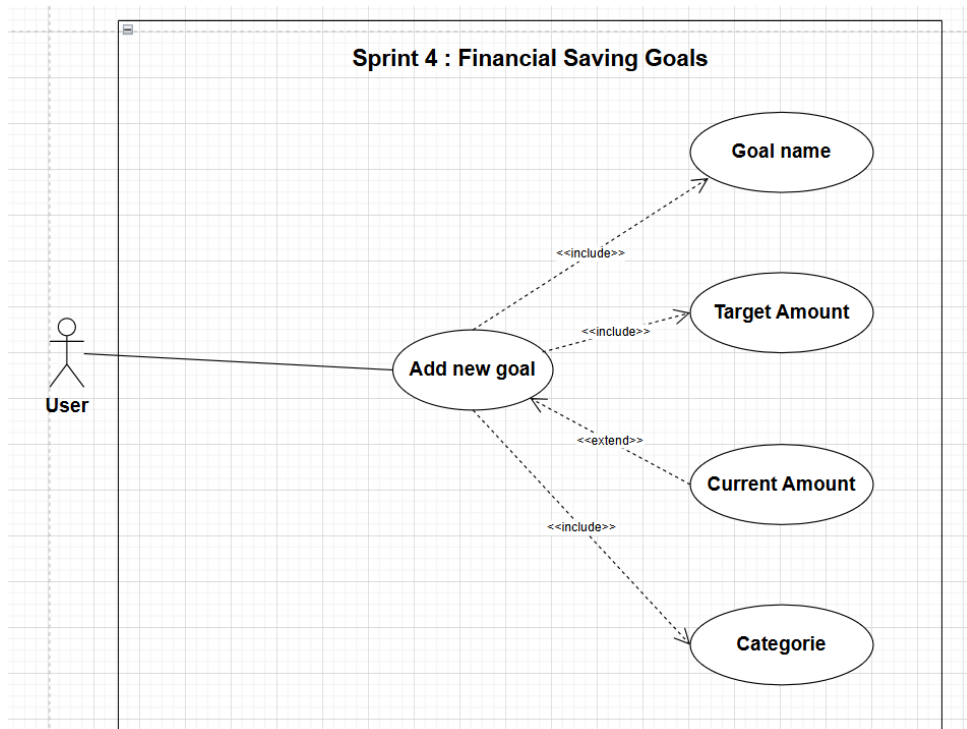


Figure 6.24: Sprint 4 Use Case Diagram

6.3.2 Use Case Description

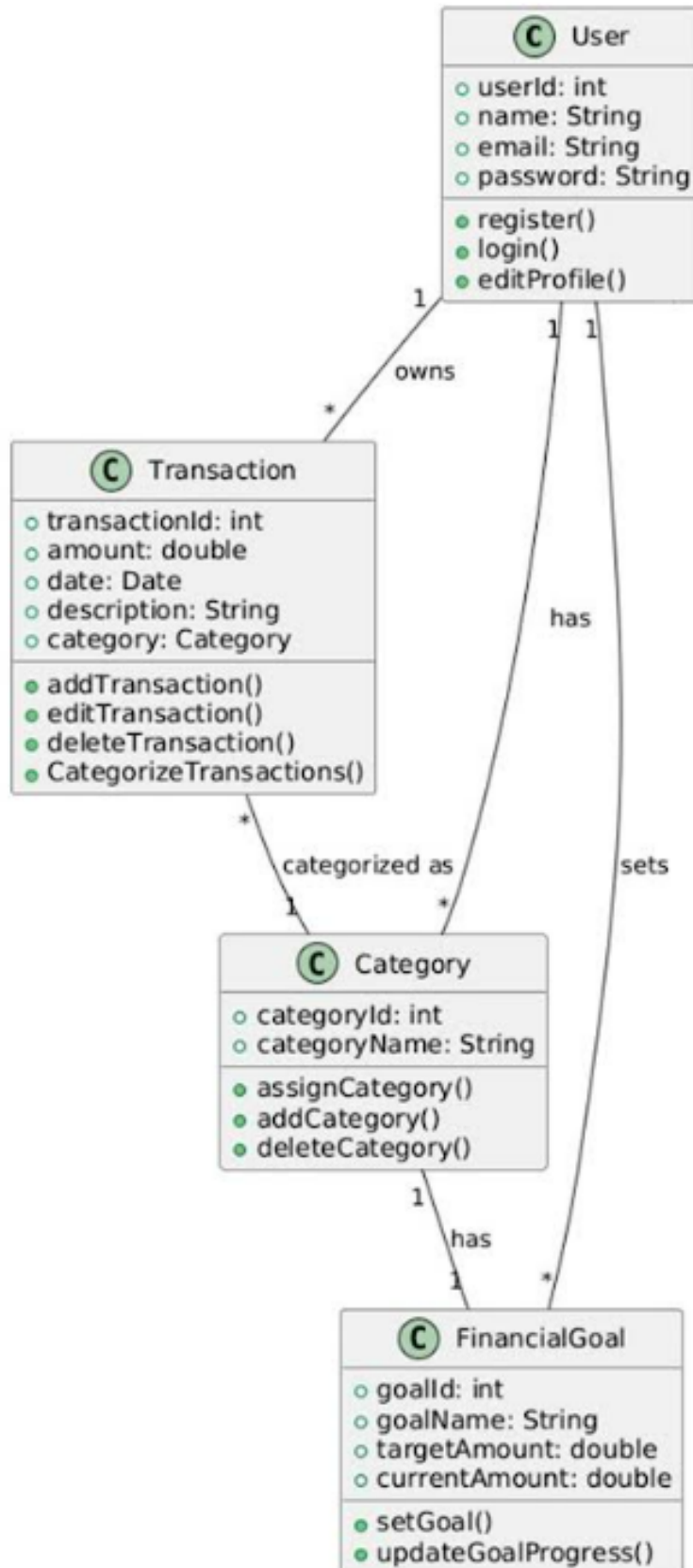
6.3.2.1 Use Case: Define Savings Goal

- **Main Actor:** Authenticated User

- **Precondition:** Be logged into the system
- **Main Scenario:**
 1. User accesses the goals section
 2. User defines a new goal (amount, date)
 3. System validates the data
 4. System records the goal
 5. System configures associated notifications
- **Postcondition:** Goal created and notifications configured

6.4 Design

6.4.1 Class Diagram



6.4.2 Detailed Sequence Diagrams

6.4.2.1 Goal Management Sequence

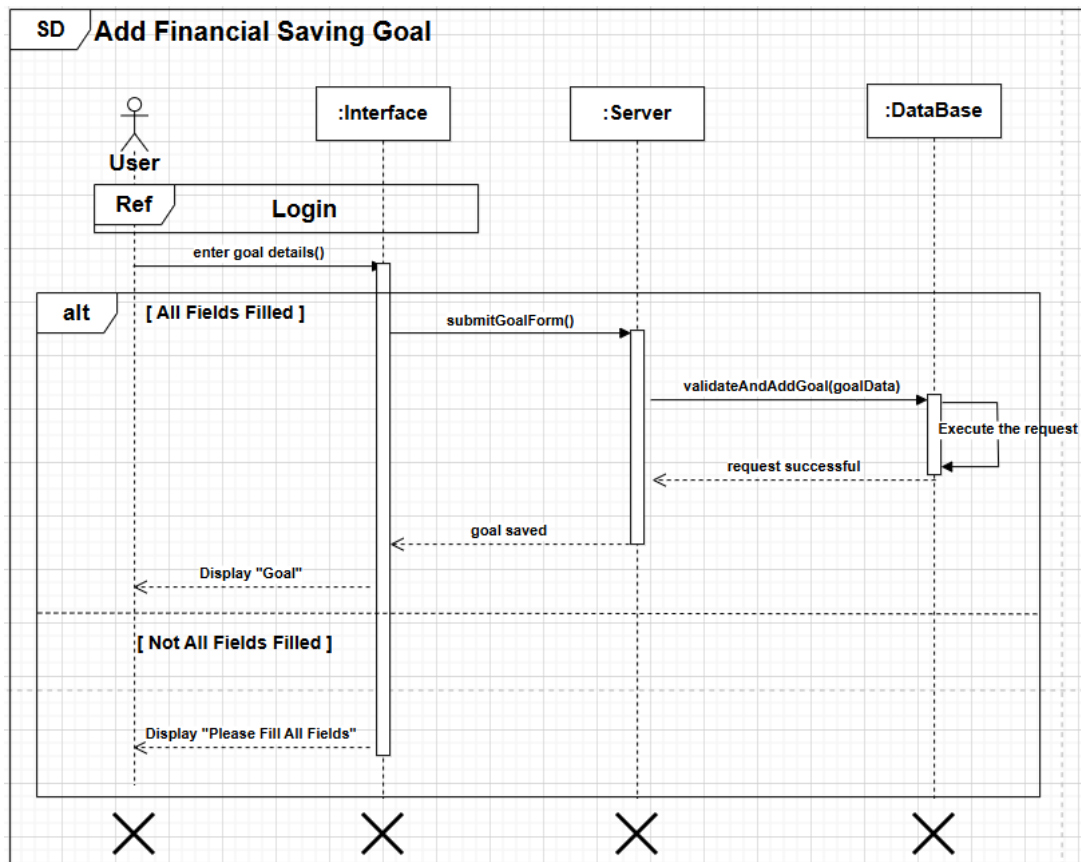


Figure 6.26: Sequence Diagram - Goal Management

6.4.2.2 Interaction Description

Goal Tracking Process:

1. User creates a savings goal
2. System records the goal
3. System regularly calculates progress
4. System sends notifications based on defined thresholds
5. User can view their progress

6.5 Implementation

This section presents the interfaces developed for goal management.

6.5.1 Goal Creation Interface

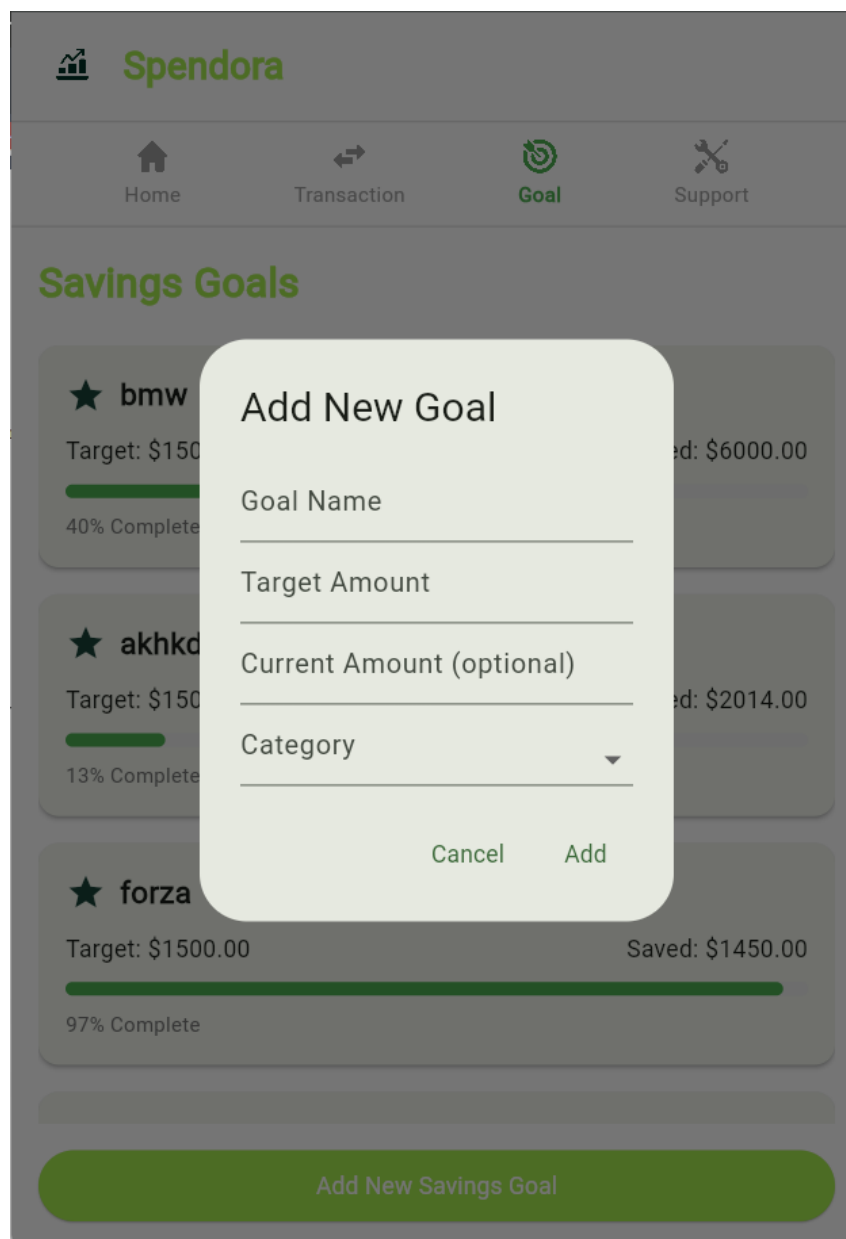


Figure 6.27: Goal Creation Interface

6.5.2 Goal Tracking Interface

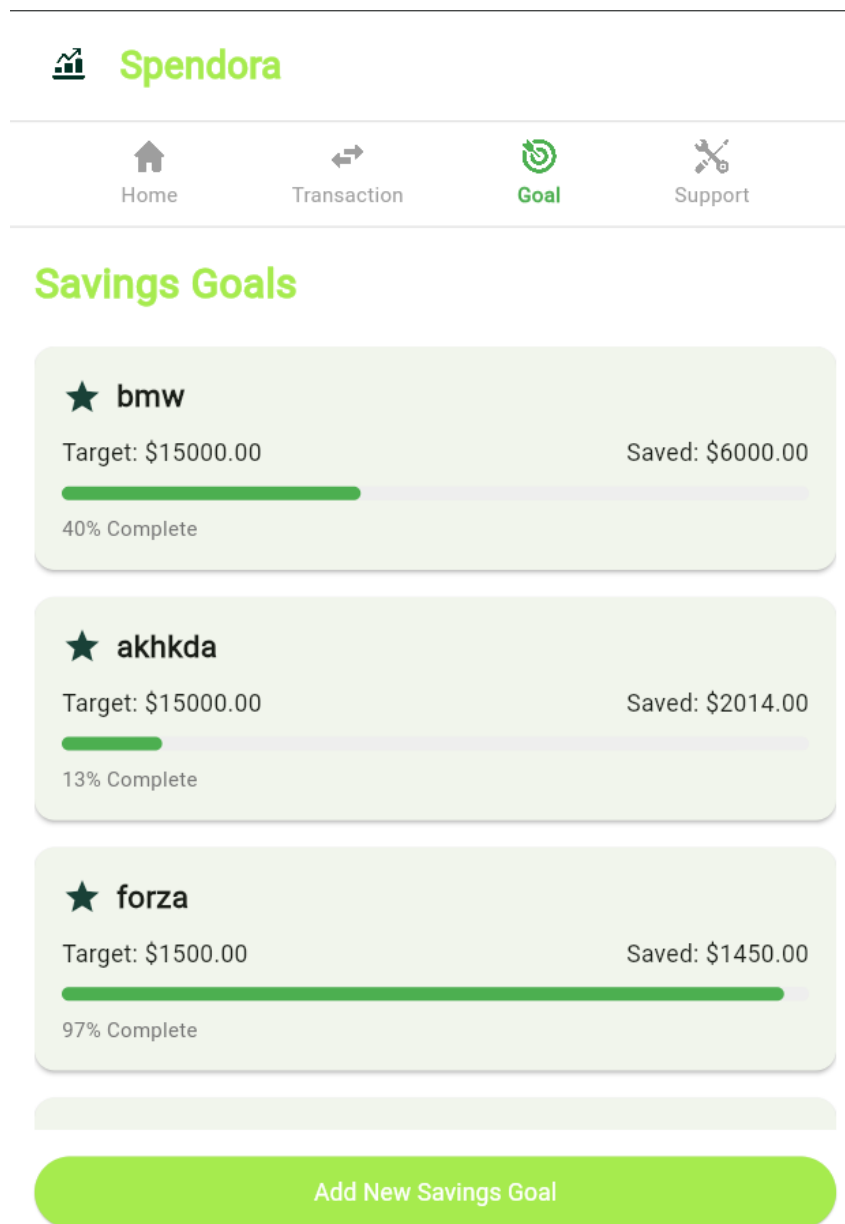


Figure 6.28: Goal Tracking Interface

6.6 Conclusion

This fourth sprint has implemented essential features to help users achieve their financial goals. The main achievements include:

- A robust savings goal management system
- An intuitive progress tracking mechanism
- A customizable notification system
- Engaging user interfaces

These features strengthen Spendora's motivational aspect, encouraging users to maintain financial discipline while providing regular feedback on their progress. User testing has shown strong appreciation for these new features, particularly for their encouraging aspect and ease of use.

Chapter 7

« Study and Implementation of Sprint 5: Reports and Export »

7.1 Introduction

This chapter presents the implementation of the fifth sprint, focused on generating financial reports and export functionality. These features allow users to analyze their financial data in depth and share it easily.

7.2 Sprint 5 Backlog

ID	User Story	Priority	Points
US17	Generate monthly financial reports	High	8
US18	Export transactions to CSV format	Medium	5
US19	Customize reports	Medium	5
US20	Schedule automatic report sending	Low	3

Table 7.12: Sprint 5 Backlog

7.3 Functional Specification

7.3.1 Use Case Diagram

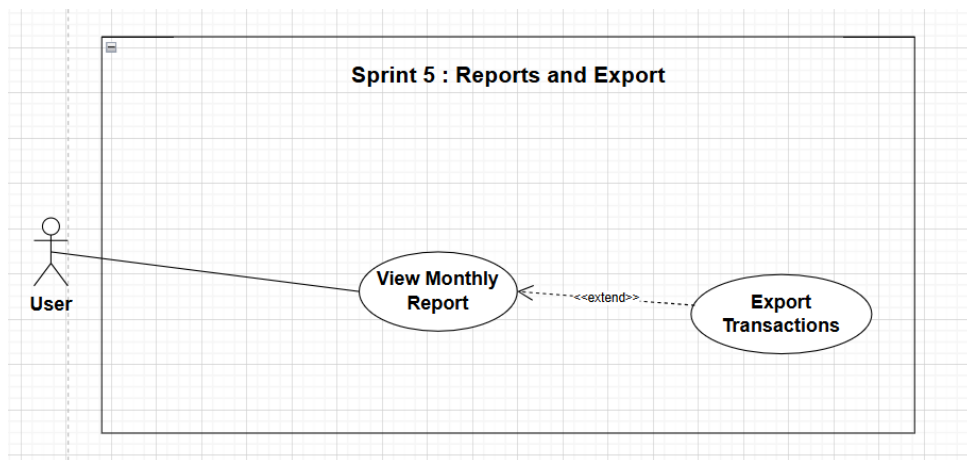


Figure 7.29: Sprint 5 Use Case Diagram

7.3.2 Use Case Description

7.3.2.1 Use Case: Generate Report

- **Main Actor:** Authenticated User
- **Precondition:** Have recorded transactions
- **Main Scenario:**
 1. User selects report period
 2. User chooses report format

3. System collects data
4. System generates report
5. System offers download

- **Postcondition:** Report generated and available

7.4 Design

7.4.1 Class Diagram

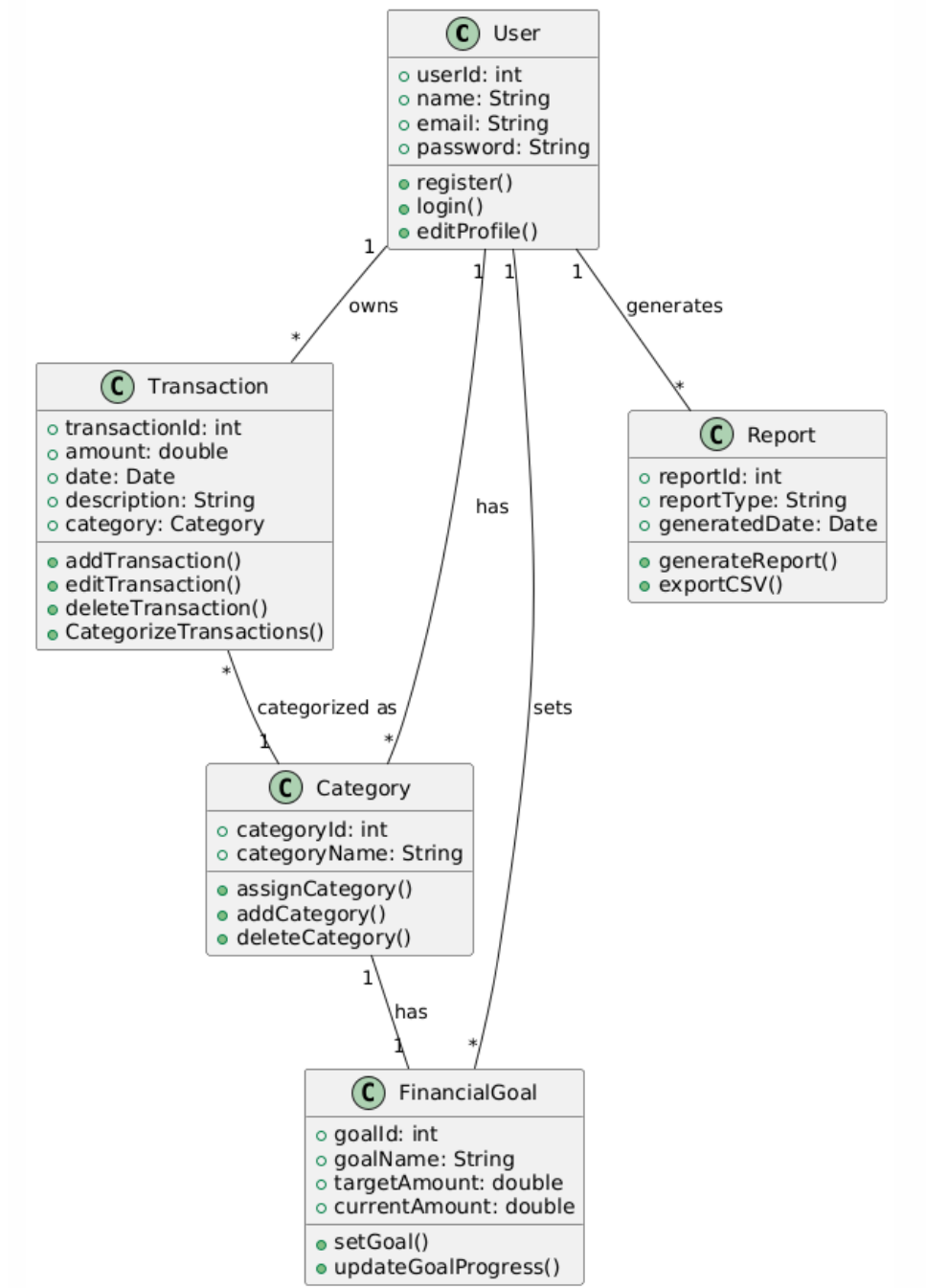


Figure 7.30: Sprint 5 Class Diagram

7.4.2 Detailed Sequence Diagrams

7.4.2.1 Report Generation Sequence

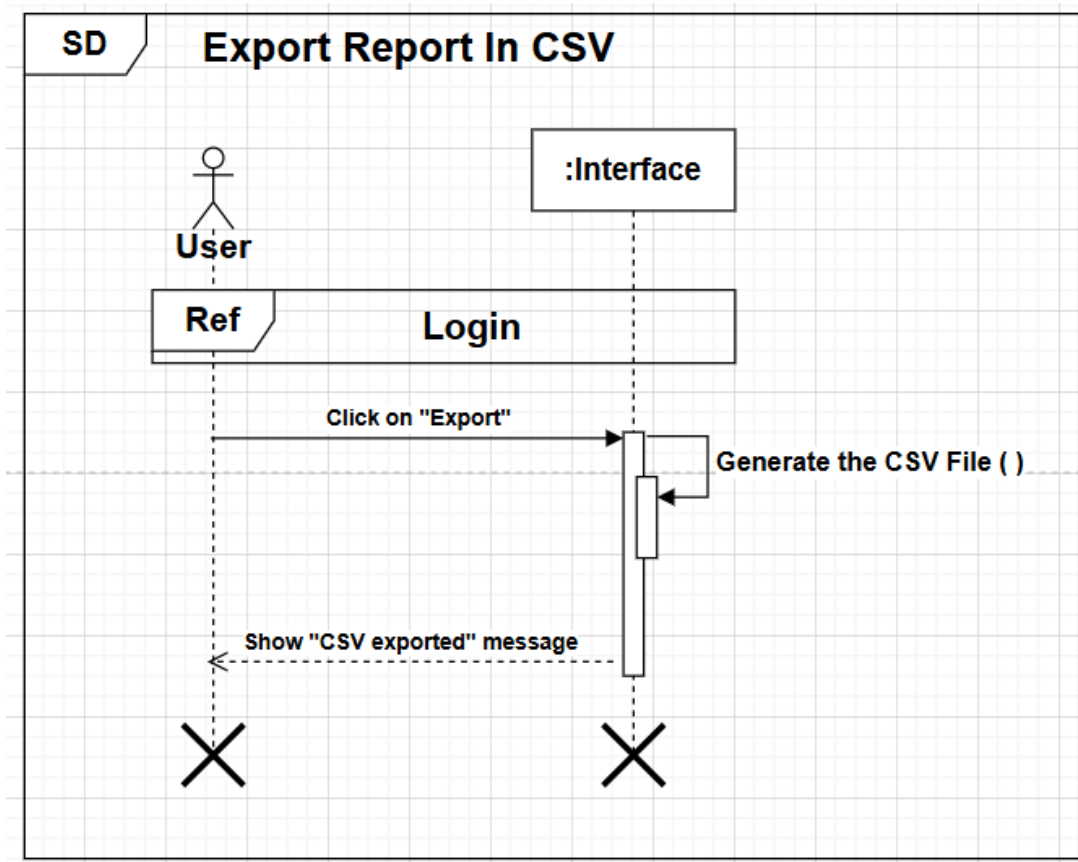


Figure 7.31: Sequence Diagram - Report Generation

7.4.2.2 Interaction Description

Report Generation Process:

1. User configures report settings
2. System retrieves necessary data
3. System applies filters and calculations
4. System formats the report according to the template
5. System generates the final file
6. User can download the report

7.5 Implementation

This section presents the interfaces developed for report management.

7.5.1 Report Generation Interface

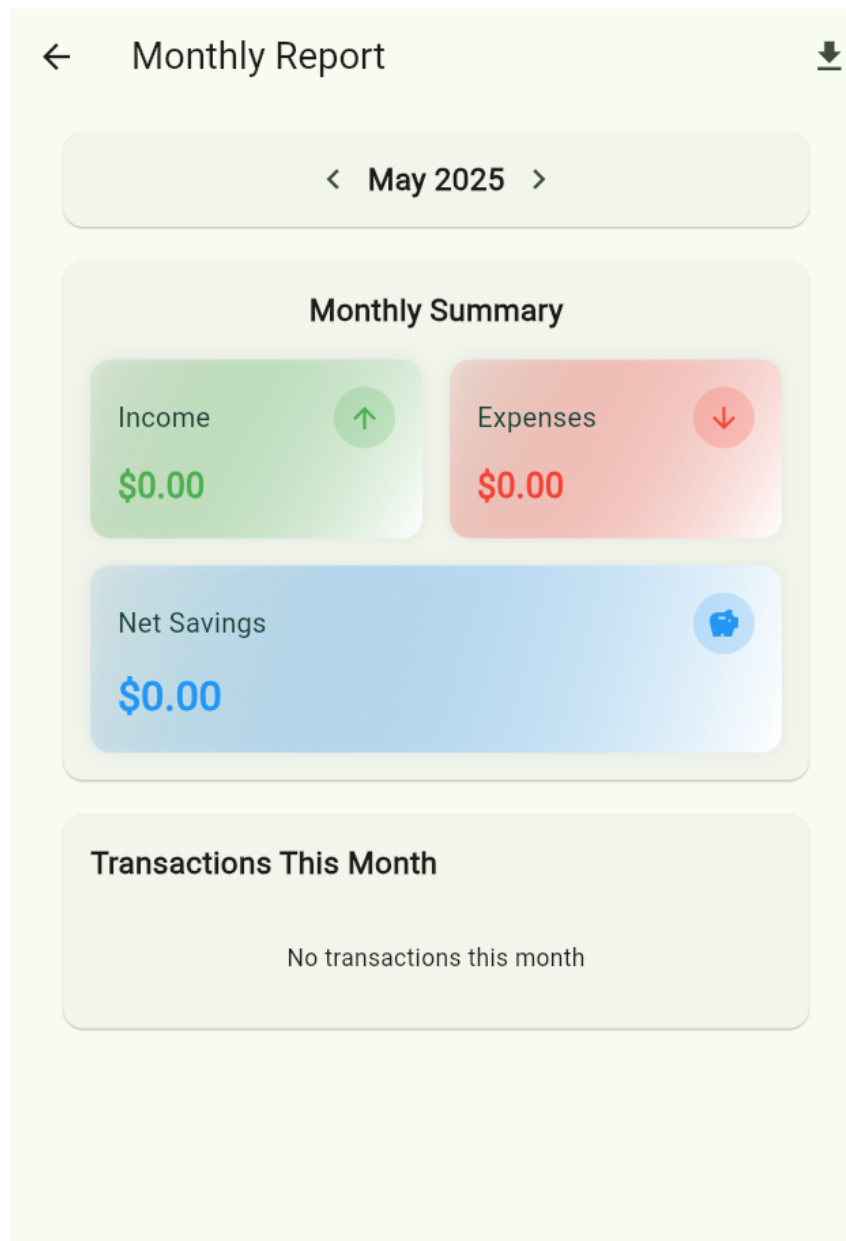


Figure 7.32: Report Generation Interface

7.6 Conclusion

This fifth sprint has enriched Spendora with advanced reporting and export features. The main achievements include:

- A flexible report generation system
- Export options to different formats
- Report customization features
- Automatic scheduling mechanism

These features meet users' needs for in-depth analysis and financial data sharing. User feedback particularly highlights the usefulness of customized reports and the ease of data export for external use.

Chapter 8

« Study and Implementation of Sprint
6:
Security and Optimization »

8.1 Introduction

This chapter presents the implementation of the sixth and final sprint, dedicated to application security and performance optimization. This final phase is crucial to ensure the protection of user data and a smooth user experience.

8.2 Sprint 6 Backlog

ID	User Story	Priority	Points
US21	Implement data encryption	High	8
US22	Add dark mode	Medium	3
US23	Optimize application performance	High	5
US24	Implement multi-currency support	Medium	5

Table 8.13: Sprint 6 Backlog

8.3 Functional Specification

8.3.1 Use Case Diagram

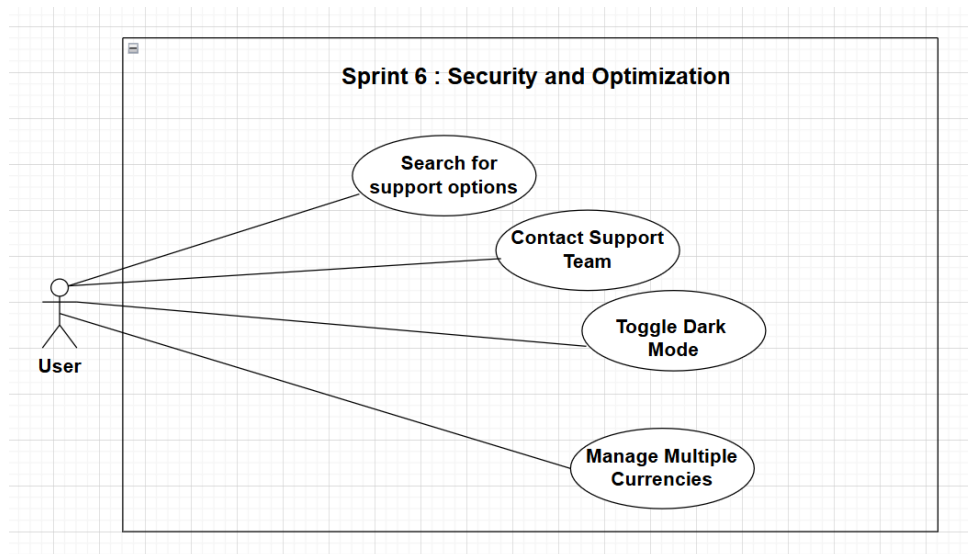


Figure 8.33: Sprint 6 Use Case Diagram

8.3.2 Use Case Description

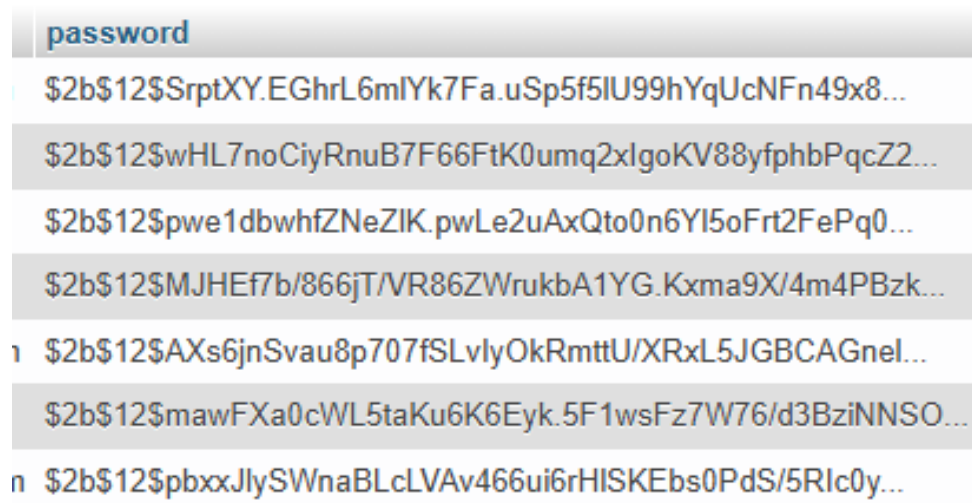
8.3.2.1 Use Case: Data Security

- **Main Actor:** System
- **Precondition:** Application running
- **Main Scenario:**

1. System encrypts sensitive data
 2. System verifies data integrity
 3. System manages sessions securely
 4. System applies security policies
- **Postcondition:** Data secured and integrity maintained

8.4 Design

8.4.1 Security Implementation



The diagram illustrates the Bcrypt password hashing process. It shows a sequence of steps where a password is transformed into a hashed representation. The password is shown in a blue box, and the resulting hash is shown in a series of grey boxes, each containing a long string of characters. The hash is composed of several parts, including a prefix, a salt, and a checksum, all concatenated together.

```

password
$2b$12$SrptXY.EGhrL6mIYk7Fa.uSp5f5IU99hYqUcNFn49x8...
$2b$12$wHL7noCiyRnuB7F66FtK0umq2xlgoKV88yfphbPqcZ2...
$2b$12$pwe1dbwhfZNeZIK.pwLe2uAxQto0n6YI5oFrt2FePq0...
$2b$12$MJHEf7b/866jT/VR86ZWrukba1YG.Kxma9X/4m4PBzk...
1 $2b$12$AXs6jnSvau8p707fSLvlyOkRmttU/XRxL5JGBCAGnel...
$2b$12$mawFXa0cWL5taKu6K6Eyk.5F1wsFz7W76/d3BziNNSO...
n $2b$12$pbxxJlySWnaBLcLVAv466ui6rHISKEbs0PdS/5Rlc0y...

```

Figure 8.34: Bcrypt Password Hashing Process

8.4.2 Security Mechanisms

- **Password Hashing:** Implementation of Bcrypt with salt rounds
- **Session Management:** Secure token-based authentication
- **Input Validation:** Server-side validation and sanitization
- **Database Security:** Protection against SQL injection
- **Activity Monitoring:** Security event logging and tracking

8.5 Implementation

8.5.1 Currency Management Interface

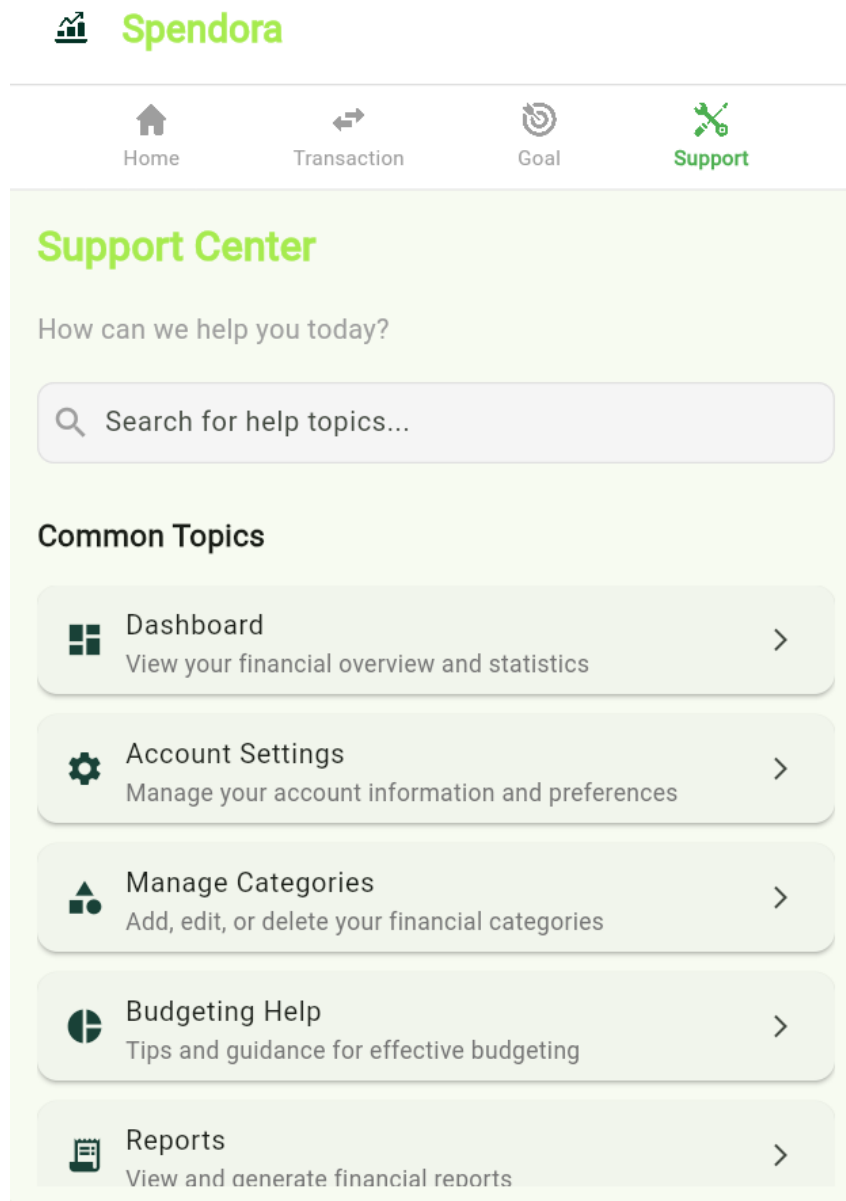


Figure 8.35: Currency Management Interface

8.6 Tests and Optimization

8.6.1 Security Tests

- Penetration testing
- Vulnerability analysis
- SQL injection testing

- Input validation testing

8.6.2 Performance Tests

- Load testing
- Response time testing
- Memory consumption testing
- Battery consumption testing

8.7 Conclusion

This final sprint has completed the Spendora application by strengthening its security and optimizing its performance. The main achievements include:

- Implementation of a robust security system
- Implementation of dark mode to improve user experience
- Optimization of general application performance
- Multi-currency support for international use

Thorough testing has demonstrated the robustness of the security mechanisms implemented and the excellent performance of the application. All features developed throughout this project now form a complete and secure personal financial management solution.

General Conclusion

This final year project allowed us to develop Spendora, a comprehensive personal finance management mobile application. Through six successive sprints, we have implemented a robust solution that meets users' needs for financial tracking and analysis.

Project Overview

The development of Spendora has led to several important achievements:

- A secure authentication system
- Complete financial transaction management
- Intuitive financial data visualizations
- A savings goals management system
- Advanced reporting features
- Secure and optimized architecture

Personal Growth

This project allowed us to:

- Master the Scrum methodology
- Deepen our skills in Flutter and Python
- Develop our expertise in application security
- Improve our teamwork capabilities
- Manage a complete mobile development project

Future Perspectives

For future versions of Spendora, several areas of improvement are possible:

- Integration with banking services
- Advanced budget planning features

- Artificial intelligence for expense analysis
- Web version of the application
- Collaborative features for family management

Challenges Encountered

During the project, we faced several challenges:

- Management of financial data security
- Performance optimization with large volumes of data
- Integration of different technologies
- Meeting development deadlines

Closing Words

The development of Spendora has been an enriching experience that allowed us to put our knowledge into practice while tackling stimulating technical challenges. The application meets the initially set objectives and provides a solid foundation for future improvements.

The skills acquired during this project, both technical and organizational, constitute a valuable asset for our future professional career. We are proud to have created a solution that can truly help users better manage their personal finances.